Model Question Papers 2026 — Complete Solutions

Tribhuvan University · Faculty of Management · BBA Eighth Semester

Paper	Course	Full Marks
1	BNK 221: Corporate Financing Decision	100
2	BNK 209: Insurance and Risk Management	100
3	BNK 202: Financial Derivatives	100

Every question (Group A to Group D, including all optional questions) is reproduced with its question text followed by a full solution. Numerical questions show step-by-step working and end with an **Exam Hint** box listing the formula to remember and the common trap.

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Paper 1 — BNK 221: Corporate Financing Decision

Tribhuvan University · Faculty of Management · BBA 8th Semester · Model Question 2026 · Full Marks 100 · Time 3 Hrs

Group A — Brief Answer Questions (10 × 2 = 20)

Q1. What do you mean by corporate financing decision?

Solution:

Corporate financing decision is the decision about *how* a firm raises the funds needed for its investments — i.e. the choice of sources, amount, mix and timing of long-term and short-term funds so that the cost of capital is minimised and the value of the firm is maximised.

Key points:

- **Capital structure decision:** proportion of debt, preferred stock and equity (financing mix).
- **Choice of instruments:** shares, debentures, term loans, leases, convertibles, warrants, short-term credit, etc.
- **Maturity mix:** balance of long-term and short-term financing (matching principle).
- **Timing and method of issue:** when and how to raise funds (public issue, rights, private placement), considering market conditions, risk, control and flexibility.

Q2. Define preemptive right.

Solution:

Preemptive right is the right of existing common shareholders to purchase any new issue of common stock (or securities convertible into it) **in proportion to their existing holding before it is offered to outsiders**.

Key points:

- It protects shareholders against **dilution of control** (voting power).
- It protects against **dilution of value**, since new shares cannot be sold to outsiders at a low price to the detriment of old shareholders.
- It is exercised through a **rights offering**; each old share receives one right.
- Rights can be exercised, sold in the market, or allowed to lapse.

Q3. List out the major types leases.

Solution:

A **lease** is a contract in which the owner of an asset (lessor) gives the right to use the asset to another party (lessee) in return for periodic rental payments. Major types:

1. **Operating (service) lease** – short-term, cancellable, lessor bears maintenance and obsolescence risk.

2. **Financial (capital) lease** – long-term, non-cancellable, fully amortized; lessee bears maintenance and risk.
3. **Sale-and-leaseback** – firm sells its own asset to a lessor and immediately leases it back.
4. **Leveraged lease** – a three-party lease in which the lessor finances a large part of the asset cost with borrowed funds from a lender.
5. **Direct lease** – lessee acquires the use of an asset it did not own before, directly from manufacturer/leasing company.

Q4. Write the meaning of trust-receipt loan.

Solution:

A **trust-receipt loan** is a form of **secured short-term inventory financing** in which the borrower holds the financed goods **in trust for the lender**. The lender (bank/finance company) pays for the goods; the borrower signs a trust receipt acknowledging that the goods are held in trust and that the **sale proceeds will be remitted to the lender** immediately when the goods are sold.

Key points:

- Used for high-value, identifiable items (automobiles, appliances, equipment) — often called **floor planning**.
- Each item is identified by serial number in the trust receipt.
- Title (security interest) remains with the lender until the loan is repaid.

Q5. Define financial distress.

Solution:

Financial distress is a situation in which a firm's operating cash flows are **not sufficient to meet its current obligations** (such as interest and principal payments), forcing it to take corrective action.

Key points:

- It ranges from **technical insolvency** (unable to pay bills as they fall due) to **legal bankruptcy** (liabilities exceed assets).
- It is mainly caused by excessive use of debt (high financial leverage) together with business risk.
- **Costs of financial distress:** direct costs (legal, administrative costs of bankruptcy) and indirect costs (lost sales, loss of key employees, stricter supplier terms, underinvestment).
- These costs limit the amount of debt in the trade-off theory of capital structure.

Q6. What is meant by call feature of bond?

Solution:

A **call feature (call provision)** gives the **issuing company the right to redeem (buy back) the bond before maturity** at a pre-specified price called the **call price**, usually par value plus a **call premium**.

Key points:

- Firms call bonds mainly when **interest rates fall**, to refund at a lower coupon (bond refunding).
- It also allows removal of restrictive covenants and forcing conversion of convertibles.
- It is disadvantageous to investors, so callable bonds carry a **higher coupon**.
- There is often a **call-protection (deferred call) period** during which the bond cannot be called.

Q7. Write about founders and angels in financing.

Solution:

Founders and angels are the earliest sources of equity for a start-up business.

- **Founders' financing:** the entrepreneur(s) invest their own savings, along with funds from family and friends ("bootstrapping"). It shows commitment and keeps full control, but the amount is limited.
- **Angel investors (business angels):** wealthy individuals who invest their personal money in early-stage (seed) ventures in exchange for equity or convertible debt.
- Angels usually invest **before venture capital**, take high risk, and often provide **advice, mentoring and network contacts** besides money.
- They expect high returns through a later exit (VC round, IPO or sale of the firm).

Q8. What do you mean by moratorium period in a term loan?

Solution:

Moratorium period (grace period) in a term loan is the **initial period after disbursement during which the borrower is not required to repay the principal** (and sometimes not even the interest).

Key points:

- It is granted because a new project needs time (construction, trial run, gestation) before it generates cash flows.
- Interest during the moratorium may be paid, capitalized, or deferred as per the agreement.
- After the moratorium, the loan is repaid in instalments (EMI/equal principal) over the remaining term.
- It reduces the early cash-flow burden and improves the project's repayment capacity.

Q9. Star Traders buys under terms of 3/10 net 50. Star Trader does not take discount and pays on 60th day. What is the annual percentage cost of stretching accounts payable? Assume 360 days in a year.

Solution:

Given:

- Credit terms = 3/10, net 50 → discount (d) = 3%, discount period = 10 days, net period = 50 days
- Actual payment day (stretched) = 60th day
- Days in a year = 360

Formula:

Annual cost of not taking discount = (d ÷ (100 − d)) × (360 ÷ (Days paid − Discount period))

Step-by-step calculation:

1. Cost per period = 3 ÷ (100 − 3) = 3 ÷ 97 = 0.030928 = 3.0928%
2. Extra days of credit obtained = 60 − 10 = 50 days
3. Number of such periods in a year = 360 ÷ 50 = 7.2
4. Annual percentage cost = 3.0928% × 7.2 = **22.27%**

(For comparison, if paid on the 50th day: (3/97) × (360/40) = 27.84%. Stretching lowers the cost.)

Answer / Decision: The annual percentage cost of stretching accounts payable to 60 days is **22.27%**.

> 💡 **Exam Hint (remember):** Cost = (d/(100−d)) × (360/(Payment day − Discount days)). Trap: use the *actual* payment day (60), not the net period (50), and divide by (100 − d), not 100. Shortcut: 3/97 ≈ 3.09%; × 360/50 = 7.2 → ≈ 22.27%.

Q10. Shikhar Company has 500,000 shares outstanding and the firm's charter provides for a cumulative voting procedure. The company has nine directors. What is the minimum number of shares needed to ensure the election of one director?

Solution:

Given:

- Total shares outstanding (S) = 500,000
- Total directors to be elected (N) = 9
- Desired directors (r) = 1

Formula:

Shares required = (r × S) ÷ (N + 1) + 1

Step-by-step calculation:

1. r × S = 1 × 500,000 = 500,000
2. N + 1 = 9 + 1 = 10
3. 500,000 ÷ 10 = 50,000
4. Shares required = 50,000 + 1 = **50,001 shares**

Answer / Decision: A minimum of **50,001 shares** is needed to ensure the election of one director.

> 💡 **Exam Hint (remember):** Shares needed = (r × S)/(N + 1) + 1. Trap: forgetting the "+1" in the denominator (N+1) or the final "+1" share. Shortcut: for one director, simply S/(N+1) + 1.

Group B — Descriptive Answer Questions (Any six, 6 × 5 = 30)

Q11. Describe the features of optimal capital structure.

Solution:

Optimal capital structure is the mix of debt, preferred stock and common equity that **minimises the weighted average cost of capital (WACC) and maximises the market value of the firm and its share price**. A sound (optimal) capital structure should have the following features:

1. **Profitability (Return):** It should make maximum use of financial leverage at minimum cost so that EPS and return on equity are maximised. Cheaper debt should be used as long as return on assets exceeds the cost of debt.
2. **Solvency / Conservatism (Risk):** Debt should not be used to the extent that it threatens the firm's solvency. The firm's cash flows must be able to cover fixed charges (interest, principal, preferred dividends) comfortably — measured by interest coverage and debt-service coverage ratios.
3. **Flexibility:** The structure should allow the firm to raise additional funds or reduce funds when needed, e.g. by keeping unused borrowing capacity, using callable bonds, and avoiding restrictive covenants.
4. **Control:** It should minimise dilution of control of existing shareholders. Excessive issue of equity dilutes control, whereas debt does not carry voting rights.
5. **Minimum cost of capital / Maximum value:** The mix should result in the lowest overall cost of capital and highest firm value.
6. **Simplicity and capacity:** The structure should be simple to manage and within the debt capacity of the firm; it should not involve too many types of securities.
7. **Matching and suitability:** It should suit the nature of the business, stability of earnings, asset structure (tangible assets support more debt) and market conditions.

Conclusion: An optimal capital structure balances return and risk — using enough debt to gain tax and leverage benefits but not so much that financial distress costs offset them.

Q12. Discuss the merits and demerits of long-term debt as a source of financing.

Solution:

Long-term debt refers to borrowing with maturity of more than one year — debentures/bonds and term loans — carrying a fixed interest obligation and repayment of principal.

Merits (from the firm's/shareholders' viewpoint):

1. **Lower cost:** Debt is less risky for investors, so the required return is lower than equity.
2. **Tax advantage:** Interest is tax-deductible, so the after-tax cost = kd(1 − T) is further reduced; it creates an interest tax shield.
3. **No dilution of control:** Debt holders have no voting rights, so ownership control remains with existing shareholders.

4. **Financial leverage:** If return on investment exceeds the cost of debt, EPS and ROE increase (trading on equity).
5. **Fixed obligation:** Debt holders do not share in excess profits; in inflation, repayment is made in cheaper money.
6. **Flexibility:** Callable debt can be retired when not needed or when rates fall.

Demerits:

1. **Fixed charge burden:** Interest and principal must be paid regardless of earnings; failure leads to default.
2. **Increased financial risk:** Higher debt raises the variability of EPS and the risk of financial distress and bankruptcy, increasing the cost of equity.
3. **Restrictive covenants:** Indentures may restrict dividends, additional borrowing, sale of assets, and working capital.
4. **Maturity/refinancing risk:** Principal must be repaid on maturity, possibly in unfavourable market conditions.
5. **Collateral requirement:** Secured debt ties up assets and reduces future borrowing capacity.
6. **Limited capacity:** A firm can borrow only up to a certain limit; beyond it, cost rises sharply.

Conclusion: Long-term debt is beneficial for firms with stable earnings and tangible assets, but it must be used within prudent limits.

Q13. Explain the concept and importance of venture capital financing.

Solution:

Concept: Venture capital (VC) is equity or equity-linked long-term financing provided by professional investors (venture capital firms/funds) to **new, young, high-risk and high-growth potential businesses**, usually unlisted, that cannot obtain funds from conventional sources. The VC takes an ownership stake, participates actively in management (board seat, advice) and exits after 5–7 years through an IPO, trade sale or buy-back, earning a high capital gain.

Stages of VC financing: seed/start-up, first stage (early), second stage (expansion), mezzanine/bridge financing, and buyout financing.

Importance of venture capital:

1. **Financing for start-ups:** Provides funds to innovative firms that lack track record and collateral and cannot get bank loans or access the stock market.
2. **Promotes innovation and technology:** Supports new products, technology and research-based ventures.
3. **Value-added services:** VCs offer managerial expertise, strategic guidance, contacts, and help in recruitment and marketing.
4. **Risk sharing:** As equity holders, VCs share the business risk; no fixed repayment burden on the young firm.
5. **Entrepreneurship and employment:** Encourages entrepreneurs, creates employment and contributes to economic growth.
6. **Path to capital market:** Prepares firms for an IPO, developing the capital market.
7. **High return for investors:** Gives investors an opportunity for high returns through portfolio diversification across ventures.

Nepalese context: Private equity and venture capital funds (PEVC) are now registered and regulated under SEBON's Specialized Investment Fund Regulation, supporting start-ups and SMEs.

Q14. What is leveraged buyout? Describe the characteristics of successful LBO.

Solution:

Leveraged Buyout (LBO): An LBO is the **acquisition of a company (or a division) by a small group of investors — often including existing management — financed mostly with borrowed funds** (often 70–90% debt), using the target firm's own assets and future cash flows as security. After the buyout, the firm usually becomes **privately held** (going private). The debt is repaid from operating cash flows and sale of non-core assets; investors earn profit by later re-listing (reverse LBO) or selling the firm. When managers buy the firm it is called a **Management Buyout (MBO)**.

Characteristics of a successful LBO candidate:

1. **Stable and predictable cash flows:** Needed to service heavy interest and principal payments.
2. **Low existing debt:** A firm with little debt has unused borrowing capacity to support the LBO debt.
3. **Separable, non-core assets:** Assets/divisions that can be sold quickly to repay part of the debt.
4. **Competent and committed management:** Experienced managers, often with an ownership stake, who are motivated to improve efficiency.
5. **Low capital expenditure requirements:** Cash is needed for debt service, not heavy reinvestment.
6. **Strong market position:** An established brand or market share in a mature industry with limited competition.
7. **Not high-tech / not subject to rapid change:** Mature, low-growth businesses with low technological and business risk are better candidates.
8. **Tangible assets for collateral** and potential for cost reduction or efficiency gains.
9. **Purchase at a reasonable price:** Overpaying ruins the return.

Conclusion: LBOs create value through tax savings on interest, improved management incentives and efficiency, but high leverage makes them risky if cash flows decline.

Q15. Differentiate between operating lease and financial lease.

Solution:

An **operating lease** is a short-term, cancellable lease in which the lessor retains the risks and rewards of ownership, while a **financial lease** is a long-term, non-cancellable lease that transfers substantially all risks and rewards of ownership to the lessee.

Basis	Operating Lease	Financial (Capital) Lease
Duration	Short-term; less than the economic life of the asset	Long-term; covers most of the economic life
Cancellability	Cancellable at the option of the lessee (with notice)	Non-cancellable

Basis	Operating Lease	Financial (Capital) Lease
Amortization	Not fully amortized; lessor recovers cost through several leases/resale	Fully amortized; lessor recovers full cost plus return from one lessee
Maintenance & insurance	Borne by the lessor (service lease)	Borne by the lessee (net lease)
Risk of obsolescence	Borne by the lessor	Borne by the lessee
Nature	Rental/service arrangement	Substitute for borrowing to buy the asset (debt-like)
Assets involved	Computers, vehicles, office equipment, assets subject to obsolescence	Land, buildings, heavy machinery, aircraft, ships
Ownership option	Usually no purchase option	Often has a bargain purchase or renewal option
Rental amount	Higher per period relative to cost (covers risk & service)	Lower; based on cost of asset + interest
Accounting (lessee)	Traditionally shown as rent expense (off-balance sheet)	Capitalized as asset and lease liability on balance sheet

Q16. The Mega Company has warrants outstanding that expire in three years. Each warrant entitles the holder to purchase two shares of common stock at an exercise price of Rs 150 per share. Determine theoretical value and premium over the theoretical value if the respective prices of common stock and warrants are:

a. Rs 140 per share and Rs 10 per warrant. b. Rs 160 per share and Rs 40 per warrant.

Solution:

Given:

- Shares per warrant (N) = 2
- Exercise price (E) = Rs 150 per share
- (a) Ps = Rs 140, warrant price = Rs 10
- (b) Ps = Rs 160, warrant price = Rs 40

Formula:

- Theoretical value of warrant (TV) = max[(Ps − E) × N, 0]
- Premium over theoretical value = Market price of warrant − TV
- Premium per share = Premium ÷ N; % premium = Premium ÷ TV × 100

Step-by-step calculation:

(a) Ps = Rs 140, warrant = Rs 10

1. (Ps − E) × N = (140 − 150) × 2 = −10 × 2 = −20 → negative, so TV = **Rs 0** (a warrant cannot have negative value; holder simply won't exercise).
2. Premium = 10 − 0 = **Rs 10** per warrant.
3. Premium per share = 10 ÷ 2 = Rs 5.
4. % premium over TV is not defined (TV = 0); the entire market price is premium (time value/speculative value).

(b) Ps = Rs 160, warrant = Rs 40

1. TV = (160 − 150) × 2 = 10 × 2 = **Rs 20**
2. Premium = 40 − 20 = **Rs 20** per warrant
3. Premium per share = 20 ÷ 2 = Rs 10
4. % premium over TV = 20 ÷ 20 × 100 = 100%

Case	Ps (Rs)	TV (Rs)	Market price of warrant (Rs)	Premium (Rs)	Premium per share (Rs)
a	140	0	10	10	5
b	160	20	40	20	10

Answer / Decision: (a) TV = Rs 0, premium = Rs 10; (b) TV = Rs 20, premium = Rs 20. The warrant sells above its theoretical value because of its leverage and the chance that the share price will rise before expiry (3 years remaining).

> 💡 **Exam Hint (remember):** TV = (Ps − E) × N, never below zero. Trap: forgetting to multiply by the number of shares per warrant (N = 2) or writing a negative TV. Shortcut: Premium = Warrant price − TV; if Ps < E, the whole warrant price is premium.

> **Q17. Akash Food Company (AFC) projects an increase in sales from Rs 1,200,000 to Rs 1,800,000 but it needs an additional Rs 200,000 of current assets to support this expansion. The money can be obtained from the bank at an interest rate of 12 percent, discount interest; no compensating balance is required. Alternatively, AFC can finance the expansion by no longer taking discounts, thus increasing accounts payable. AFC purchases on the terms of 3/10, net 40, but it can delay payment for an additional 20 days and thus becoming 20 days past due without a penalty.**

Based strictly on effective annual interest rate comparisons, how should AFC finance its expansion?

Solution:

Given:

- Funds needed = Rs 200,000
- Bank loan: 12% discount interest, no compensating balance
- Trade credit: 3/10, net 40; payment can be stretched 20 days → paid on day 60
- Year = 360 days

Formula:

- Discount loan: Effective rate = Interest rate ÷ (1 − Interest rate)
- Trade credit: Cost = (d ÷ (100 − d)) × (360 ÷ (Payment day − Discount period))

Step-by-step calculation:

1. Bank loan (discount interest)

- Effective rate = 0.12 ÷ (1 − 0.12) = 0.12 ÷ 0.88 = 0.13636 = **13.64%**
- (Check: to get Rs 200,000 usable, borrow 200,000 ÷ 0.88 = Rs 227,273; interest = 227,273 × 12% = Rs 27,273; 27,273 ÷ 200,000 = 13.64%.)

2. Trade credit (forgo discount, pay on day 60)

- Payment day = 40 + 20 = 60
- Cost = (3 ÷ 97) × (360 ÷ (60 − 10)) = 0.030928 × 7.2 = 0.2227 = **22.27%**
- Without stretching (paid on day 40): (3 ÷ 97) × (360 ÷ 30) = 0.030928 × 12 = 37.11%

Source	Effective annual cost
Bank loan (12% discount interest)	13.64%
Trade credit, paid on day 60 (stretched)	22.27%
Trade credit, paid on day 40 (not stretched)	37.11%

Answer / Decision: The bank loan costs **13.64%**, which is lower than the 22.27% cost of trade credit even after stretching. AFC should **finance the expansion with the bank loan** and continue taking the cash discounts.

> 💡 **Exam Hint (remember):** Discount loan rate = r/(1 − r); trade credit cost = (d/(100−d)) × 360/(pay day − discount days). Trap: using the net period (40) instead of the stretched day (60), or forgetting that discount interest raises the effective rate above 12%. Shortcut: choose the source with the lower effective rate — here 13.64% < 22.27%, so borrow.

Group C — Analytical Answer Questions (Any three, 3 × 10 = 30)

Q18. Explain the methods of issuing securities. Also discuss the process of initial public offering.

Solution:

A. Methods of issuing securities

Firms raise long-term funds from the primary market through the following methods:

1. **Public issue (general cash offer):** Securities are offered to the general public through a prospectus. - **Initial Public Offering (IPO):** the first issue of shares to the public by a company, after which it becomes listed. - **Further Public Offering (FPO) / seasoned offering:** an additional issue by an already-listed company.
2. **Rights issue (privileged subscription):** New shares are offered first to existing shareholders in proportion to their holdings, usually at a price below market, honouring the preemptive right. It is cheap and preserves control. In Nepal, banks and other listed companies widely use rights issues (e.g. 1:1, 10:3).
3. **Private placement:** Securities are sold directly to a small number of institutional investors (banks, insurance companies, pension funds, mutual funds) without a public offer. It is quick, flexible and cheaper, but the issue size is limited and investors may demand higher return.
4. **Underwritten (firm-commitment / bought deal) offering:** The underwriter (investment bank/merchant banker) buys the whole issue at a fixed price and resells it to the public, bearing the risk of unsold shares. In a **bought deal**, a single underwriter buys the entire issue quickly.

5. **Best-efforts offering:** The investment banker acts only as an agent and sells as much as possible; unsold securities are returned to the issuer. Risk remains with the issuer.
6. **Competitive bidding vs. negotiated offering:** In competitive bidding, the issuer invites bids from several underwriters and awards to the highest bidder; in a negotiated offer, the terms are negotiated with one underwriter.
7. **Book building:** The price is discovered through bids from investors within a price band; the final price is fixed based on demand. SEBON introduced book building (for qualified institutional investors) in Nepal.
8. **Shelf registration:** A firm registers a large issue once and sells portions over a period (up to two years) as needed.
9. **Other methods:** Employee share issue, preferential allotment, and offers for sale by existing promoters.

B. Process of Initial Public Offering (Nepalese context)

1. **Decision to go public:** Board and general meeting approve the IPO, amend the memorandum/articles and convert to a public limited company if needed.
2. **Appointing an issue manager:** A SEBON-licensed merchant banker (issue manager) is appointed to manage the issue; **underwriters** are appointed to guarantee subscription.
3. **Due diligence and preparation:** The issue manager examines financial statements, legal documents, projections and business plans.
4. **Credit rating:** IPO must be graded by a credit rating agency (e.g. ICRA Nepal, CARE Ratings Nepal).
5. **Preparing the prospectus and SEBON approval:** The draft prospectus with company details, objectives, risk factors and financials is submitted to the **Securities Board of Nepal (SEBON)**, which reviews and approves it.
6. **Pricing the issue:** Fixed-price at par (Rs 100) for most companies, or at a premium (approved by SEBON) / via book building for eligible firms.
7. **Marketing and announcement:** Publication of the issue opening notice; road shows/promotion; collection centres (ASBA – member banks) and online application through **MeroShare** (CDS & Clearing Ltd.).
8. **Subscription period:** Investors apply through ASBA within the open period; quotas for employees, mutual funds, project-affected locals and Nepali workers abroad are reserved.
9. **Allotment:** If oversubscribed, shares are allotted by lottery/pro-rata under SEBON rules; refunds are released.
10. **Dematerialisation and listing:** Allotted shares are credited to investors' **demat accounts at CDS and Clearing Ltd.**, and the shares are **listed on NEPSE** for trading in the secondary market.
11. **Post-issue compliance:** Reporting to SEBON/NEPSE and continuing disclosure obligations.

Conclusion: Choosing the proper method depends on the issue size, cost, urgency, control considerations and market conditions; an IPO converts a private firm into a listed public company, giving it access to a wider capital base and liquidity for its shares.

> **Q19. The Nepal Energy Company has announced a rights offer to raise Rs 600 million for a new Gas Station. The stock currently sells for Rs 600 per share, and there are 4.8 million shares outstanding. Par value is Rs 100.**

a. What is the maximum possible subscription price? What is the minimum? b. If the subscription price is set at Rs 500 per share, how many shares must be sold? c. How many rights will it take to buy one share? d. What is the value of a right? e. What is the ex-rights price? f. Show how a shareholder with 1,000 shares before the offering and no desire (or money) to buy additional shares is not harmed by the rights offer.

Solution:

Given:

- Funds to be raised = Rs 600,000,000
- Current market price (Po, rights-on) = Rs 600
- Shares outstanding (No) = 4,800,000
- Par value = Rs 100; Subscription price (S) = Rs 500

Formula:

- New shares = Funds required ÷ Subscription price
- Rights per new share (N) = Old shares ÷ New shares
- Value of right (rights-on) R = (Po − S) ÷ (N + 1)
- Ex-rights price Px = (N × Po + S) ÷ (N + 1); value of right (ex-rights) R = (Px − S) ÷ N

Step-by-step calculation:

(a) Maximum and minimum subscription price

- **Maximum = Rs 600** (the current market price). If the subscription price were above Rs 600, nobody would subscribe, since shares can be bought more cheaply in the market. (In practice it is set slightly below market to ensure success.)
- **Minimum = Rs 100** (par value). Under the Companies Act, shares cannot be issued at a discount below par value. Any price between Rs 100 and Rs 600 is possible.

(b) Number of new shares

- New shares = 600,000,000 ÷ 500 = **1,200,000 shares**

(c) Rights needed to buy one share

- N = 4,800,000 ÷ 1,200,000 = **4 rights per new share** (ratio 1 new share for every 4 held)

(d) Value of a right

- Rights-on: R = (600 − 500) ÷ (4 + 1) = 100 ÷ 5 = **Rs 20**
- Check (ex-rights): R = (580 − 500) ÷ 4 = 80 ÷ 4 = Rs 20

(e) Ex-rights price

- Px = (4 × 600 + 500) ÷ (4 + 1) = (2,400 + 500) ÷ 5 = 2,900 ÷ 5 = **Rs 580**
- Or Px = Po − R = 600 − 20 = Rs 580
- (Check by total value: (4,800,000 × 600 + 600,000,000) ÷ 6,000,000 = 3,480,000,000 ÷ 6,000,000 = Rs 580)

(f) Position of a shareholder with 1,000 shares

Position	Option 1: Sell rights	Option 2: Exercise rights	Option 3: Do nothing
Wealth before offer	1,000 × 600 = 600,000	1,000 × 600 = 600,000	600,000
Shares after offer	1,000	1,000 + 1,000/4 = 1,250	1,000
Value of shares after	1,000 × 580 = 580,000	1,250 × 580 = 725,000	580,000
Cash from sale of rights	1,000 × 20 = 20,000	—	0
Cash paid for new shares	—	250 × 500 = (125,000)	—
Net wealth after	**600,000**	725,000 − 125,000 = **600,000**	**580,000** (loss 20,000)

Answer / Decision: (a) Max Rs 600, Min Rs 100; (b) 1,200,000 shares; (c) 4 rights; (d) Rs 20; (e) Rs 580; (f) A shareholder who does not want to buy simply **sells the 1,000 rights for Rs 20,000**; share value falls to Rs 580,000 but total wealth remains **Rs 600,000** — no loss. Only a shareholder who neither exercises nor sells the rights (lets them lapse) loses Rs 20,000.

> 💡 **Exam Hint (remember):** R = (Po − S)/(N + 1) rights-on; Px = (N·Po + S)/(N + 1); R = (Px − S)/N ex-rights. Trap: using N instead of N + 1 with the rights-on price, or forgetting that the minimum price is par value. Shortcut: N = old shares ÷ new shares; Px = Po − R.

Q20. The Kathmandu Electronic Company (KEC) intends to borrow Rs 800,000 to support its short-term financing requirements during the next year. The financing alternatives offered by the bank include:

Alternative 1: A discount interest loan with a simple interest of 15 percent and no compensating balance requirement. Alternative 2: A 14 percent simple interest loan that has a 15 percent compensating balance requirement. Alternative 3: Rs 1 million revolving line of credit with simple interest of 12 percent paid on the amount borrowed and a 1 percent commitment fee. a. Compute the effective cost (rate) of each financing alternative assuming KEC borrows Rs 800,000. Which alternative should it use? b. Discuss the factors other than cost to be considered while choosing a bank for short-term loan.

Solution:

(a) Effective cost of each alternative

Given:

- Usable funds needed = Rs 800,000; period = 1 year
- Alt 1: 15% discount interest
- Alt 2: 14% simple interest, 15% compensating balance (CB)
- Alt 3: Rs 1,000,000 revolving credit, 12% on amount used, 1% commitment fee on unused portion

Formula:

- Discount loan: Effective rate = r ÷ (1 − r)

- Simple interest with CB: Effective rate = r ÷ (1 − CB)
- Revolving credit: Effective rate = (Interest + Commitment fee) ÷ Usable funds
- Amount to borrow for required usable funds = Usable funds ÷ (1 − r or CB)

Step-by-step calculation:

Alternative 1 – Discount interest loan (15%)

- Effective rate = 0.15 ÷ (1 − 0.15) = 0.15 ÷ 0.85 = **17.65%**
- To have Rs 800,000 usable: borrow = 800,000 ÷ 0.85 = Rs 941,176
- Interest = 941,176 × 15% = Rs 141,176
- Check: 141,176 ÷ 800,000 = 17.65%
- (If face value borrowed is only Rs 800,000: interest 120,000; usable 680,000; 120,000 ÷ 680,000 = 17.65% — same rate.)

Alternative 2 – 14% loan with 15% compensating balance

- Effective rate = 0.14 ÷ (1 − 0.15) = 0.14 ÷ 0.85 = **16.47%**
- To have Rs 800,000 usable: borrow = 800,000 ÷ 0.85 = Rs 941,176 (CB = 141,176)
- Interest = 941,176 × 14% = Rs 131,765
- Check: 131,765 ÷ 800,000 = 16.47%

Alternative 3 – Revolving credit (Rs 1,000,000 line)

- Interest = 12% × 800,000 = Rs 96,000
- Unused portion = 1,000,000 − 800,000 = Rs 200,000
- Commitment fee = 1% × 200,000 = Rs 2,000
- Total cost = 96,000 + 2,000 = Rs 98,000
- Effective rate = 98,000 ÷ 800,000 = **12.25%**

Alternative	Amount borrowed (Rs)	Annual cost (Rs)	Effective rate
1. 15% discount loan	941,176	141,176	17.65%
2. 14% loan, 15% CB	941,176	131,765	16.47%
3. Revolving credit 12% + 1% fee	800,000	98,000	**12.25%**

Answer / Decision: Alternative 3 (revolving line of credit) has the lowest effective cost of **12.25%**, so KEC should choose **Alternative 3**.

> 💡 **Exam Hint (remember):** Discount loan = r/(1−r); compensating balance = r/(1−CB); revolving credit = (r × used + fee% × unused)/used. Trap: charging the commitment fee on the whole Rs 1 million instead of only the unused Rs 200,000, or dividing by the gross loan instead of usable funds. Shortcut: 0.15/0.85 = 17.65%, 0.14/0.85 = 16.47%, (96,000 + 2,000)/800,000 = 12.25%.

(b) Factors other than cost in choosing a bank

1. **Willingness to assume risk:** Banks differ in their attitude towards risk; some are conservative while others lend to riskier borrowers.
2. **Advice and counsel:** A bank that provides financial advice and analytical services is valuable, especially to small firms.
3. **Loyalty to customers:** Whether the bank will support the borrower in bad times rather than calling the loan or refusing renewal.

4. **Specialization:** Banks specialise in particular industries (agriculture, hydropower, manufacturing, trade); a specialised bank understands the business better.
5. **Maximum loan size:** The bank's capital base and single-obligor limit determine how much it can lend to one customer; a bank big enough to meet growing needs should be chosen.
6. **Merchant banking and other services:** Availability of trade finance (LC, guarantees), cash management, foreign exchange, online banking, and capital market services.
7. **Stability and soundness of the bank:** A financially strong bank (adequate capital, low NPA) is more dependable.
8. **Relationship and convenience:** Long-term relationship, branch network, speed of decision and processing.
9. **Collateral and covenant requirements:** Less restrictive security and covenant terms give more flexibility.

> **Q21. Himalaya Company can lease equipment for three years, making annual payments of Rs 150,000 per year at the end of each year or they can buy the equipment for Rs 300,000. At the end of third years, the equipment will have Rs 60,000 salvage value. The firm's before tax cost of debt is 10 percent. The company uses straight-line depreciation and has a 40 percent tax rate.**

a. Calculate cost of leasing. b. Calculate cost of purchasing. Should the machine be leased or purchased? c. The appropriate discount rate for cash flows used in the analysis is the firm's after-tax cost of debt. Why?

Solution:

Given:

- Lease payment (L) = Rs 150,000 at end of each year, n = 3 years
- Purchase cost = Rs 300,000; Salvage value (SV) = Rs 60,000 at end of year 3
- Before-tax cost of debt (kd) = 10%; Tax rate (T) = 40%
- Straight-line depreciation

Formula:

- Discount rate = $kd(1 - T)$
- PV of leasing = $L(1 - T) \times PVIFA(kd(1-T), n)$
- Depreciation = (Cost − Salvage) ÷ n
- PV cost of purchasing = Cost − PV of depreciation tax shield − PV of salvage
- Net Advantage of Leasing (NAL) = PV cost of purchasing − PV cost of leasing

Step-by-step calculation:

Discount rate: $kd(1 - T) = 10\% \times (1 - 0.40) =$ **6%**

PV factors at 6%: PVIF(6%,1) = 0.9434; PVIF(6%,2) = 0.8900; PVIF(6%,3) = 0.8396; PVIFA(6%,3) = 2.6730

(a) Cost of leasing

1. After-tax lease payment = 150,000 × (1 − 0.40) = Rs 90,000 per year
2. PV = 90,000 × PVIFA(6%,3) = 90,000 × 2.6730 = **Rs 240,570**

Year	Lease payment	Tax saving (40%)	After-tax payment	PVIF @6%	PV (Rs)
1	150,000	60,000	90,000	0.9434	84,906
2	150,000	60,000	90,000	0.8900	80,100
3	150,000	60,000	90,000	0.8396	75,564
				Total	**240,570**

(b) Cost of purchasing

Assumption: Straight-line depreciation is charged on the depreciable base net of salvage, so book value at end of year 3 = salvage value and there is no tax on sale.

1. Annual depreciation = (300,000 − 60,000) ÷ 3 = 240,000 ÷ 3 = Rs 80,000
2. Depreciation tax shield = 80,000 × 0.40 = Rs 32,000 per year
3. PV of tax shield = 32,000 × 2.6730 = Rs 85,536
4. PV of salvage = 60,000 × 0.8396 = Rs 50,376
5. PV cost of purchasing = 300,000 − 85,536 − 50,376 = **Rs 164,088**

Year	Cash outflow / (inflow)	Amount (Rs)	PVIF @6%	PV (Rs)
0	Purchase cost	300,000	1.0000	300,000
1	Depreciation tax shield	(32,000)	0.9434	(30,189)
2	Depreciation tax shield	(32,000)	0.8900	(28,480)
3	Depreciation tax shield	(32,000)	0.8396	(26,867)
3	Salvage value	(60,000)	0.8396	(50,376)
	Net PV cost of purchasing			**164,088**

(Rounding year-wise PVs gives 164,088.)

Net Advantage of Leasing (NAL) = PV cost of purchasing − PV cost of leasing = 164,088 − 240,570 = **− Rs 76,482**

Note (alternative assumption): If depreciation is charged on the full Rs 300,000 (Rs 100,000 per year), tax shield = 40,000 × 2.6730 = 106,920; the salvage Rs 60,000 is then fully taxable, after-tax salvage = 36,000, PV = 36,000 × 0.8396 = 30,226; cost of purchasing = 300,000 − 106,920 − 30,226 = Rs 162,854. The decision (buy) does not change.

Answer / Decision: PV cost of leasing = **Rs 240,570**; PV cost of purchasing = **Rs 164,088**. Since purchasing is cheaper by Rs 76,482 (NAL is negative), Himalaya Company should **purchase (borrow and buy) the equipment**, not lease it.

(c) Why the after-tax cost of debt is the discount rate

1. **Lease is a substitute for debt:** A financial lease is a contractual, fixed obligation just like debt-service payments; failure to pay leads to loss of the asset or default. Leasing displaces the firm's debt capacity, so it should be compared with borrowing to buy.
2. **Low risk of the cash flows:** Lease payments, depreciation tax shields and the purchase price are known with (near) certainty, so they have about the same risk as the firm's debt — not the risk of the firm's operating cash flows. Hence WACC (which includes equity risk) is not appropriate.
3. **After-tax flows require an after-tax rate:** All cash flows in the analysis are after-tax (lease payments are deductible, interest is deductible), so consistency requires discounting at kd(1 − T).

4. **Opportunity cost of financing:** The firm's alternative to leasing is a secured loan; its after-tax cost is the true opportunity cost of the funds. (Salvage value is riskier, and some analysts discount it at a higher rate such as WACC.)

> 💡 **Exam Hint (remember):** Discount rate = kd(1−T); PV lease = L(1−T) × PVIFA; PV buy = Cost − (Dep × T × PVIFA) − SV × PVIF; NAL = PV buy − PV lease. Trap: discounting at 10% (pre-tax) or forgetting the tax shield on lease payments and depreciation. Shortcut: choose the lower PV cost — positive NAL → lease, negative NAL → buy.

Group D — Comprehensive / Case Study (20)

Q22. Read the following case carefully and answer the question that follow:

The Himalayan Herbal Company (HHC) has grown rapidly during the past five years. Recently, it has discovered new investment opportunity. Company is planning to increase debt to finance the new project. There are two alternatives to raise debt capital for the company: i. convertible debenture, ii. bonds with warrants. HH Company is contemplating raising Rs 10 million by means of a debt issue. It has the following alternatives:

a. A 20-year, 8 percent convertible debenture issue with a Rs 50 conversion price and Rs 1,000 face value. b. A 20-year, 10 percent straight bond issue. Each Rs 1,000 bond has a detachable warrant to purchase four shares for Rs 50 a share.

The company has a 50 percent tax rate, and its stock is currently selling at Rs 40 a share. Its earnings before interest and taxes are a constant 20 percent of its total capitalization, which currently appears as follows:

Item	Amount
Common stock (par Rs 5)	Rs 5,000,000
Additional paid-in capital	10,000,000
Retained earnings	15,000,000
Total	Rs 30,000,000

i. Show the capitalizations before and after conversion of convertible. ii. Show the capitalizations before and after exercise of warrants. iii. Compute earnings per share currently and before and after conversion of convertible and exercise of warrant. iv. Describe the reasons for issuing warrants and convertible.

Solution:

Given:

- Debt issue = Rs 10,000,000 → number of Rs 1,000 bonds = 10,000,000 ÷ 1,000 = 10,000
- Shares outstanding = 5,000,000 ÷ 5 = **1,000,000 shares**
- Convertible: 8% coupon, conversion price Rs 50
- Bonds with warrants: 10% coupon, 4 warrants (shares) per bond at Rs 50
- EBIT = 20% of total capitalization; Tax = 50%

Formula:

- Conversion ratio = Face value ÷ Conversion price
- New shares on conversion = Number of bonds × Conversion ratio
- New shares on exercise = Number of bonds × Shares per warrant; Cash received = New shares × Exercise price
- Increase in common stock = New shares × Par; Increase in APIC = New shares × (Issue price − Par)
- EPS = (EBIT − Interest)(1 − T) ÷ Number of shares

(i) Capitalization — convertible debentures

Step-by-step calculation:

1. Conversion ratio = 1,000 ÷ 50 = 20 shares per debenture
2. New shares = 10,000 × 20 = 200,000 shares
3. Increase in common stock = 200,000 × Rs 5 = Rs 1,000,000
4. Increase in APIC = 200,000 × (50 − 5) = 200,000 × 45 = Rs 9,000,000 (total Rs 10,000,000 = debt converted)

Item	Before conversion (Rs)	After conversion (Rs)
Convertible debentures (8%)	10,000,000	—
Common stock (par Rs 5)	5,000,000	6,000,000
Additional paid-in capital	10,000,000	19,000,000
Retained earnings	15,000,000	15,000,000
Total capitalization	**40,000,000**	**40,000,000**
Shares outstanding	1,000,000	1,200,000

Conversion merely transfers Rs 10 million from debt to equity; total capitalization is unchanged.

(ii) Capitalization — bonds with warrants

Step-by-step calculation:

1. New shares on exercise = 10,000 bonds × 4 = 40,000 shares
2. Cash received = 40,000 × Rs 50 = Rs 2,000,000
3. Increase in common stock = 40,000 × 5 = Rs 200,000
4. Increase in APIC = 40,000 × (50 − 5) = 40,000 × 45 = Rs 1,800,000

Item	Before exercise (Rs)	After exercise (Rs)
Bonds (10%)	10,000,000	10,000,000
Common stock (par Rs 5)	5,000,000	5,200,000
Additional paid-in capital	10,000,000	11,800,000
Retained earnings	15,000,000	15,000,000
Total capitalization	**40,000,000**	**42,000,000**
Shares outstanding	1,000,000	1,040,000

Warrants are detachable, so the bonds remain outstanding after exercise and new cash of Rs 2 million increases total capitalization.

(iii) Earnings per share

Step-by-step calculation:

- Currently: EBIT = 20% × 30,000,000 = Rs 6,000,000
- Convertible (before and after): EBIT = 20% × 40,000,000 = Rs 8,000,000; interest before conversion = 8% × 10,000,000 = Rs 800,000
- Warrants before exercise: EBIT = Rs 8,000,000; interest = 10% × 10,000,000 = Rs 1,000,000
- Warrants after exercise: EBIT = 20% × 42,000,000 = Rs 8,400,000; interest = Rs 1,000,000

Particulars	Currently	Convertible: before conversion	Convertible: after conversion	Warrants: before exercise	Warrants: after exercise
Total capitalization	30,000,000	40,000,000	40,000,000	40,000,000	42,000,000
EBIT (20%)	6,000,000	8,000,000	8,000,000	8,000,000	8,400,000
Less: Interest	0	800,000	0	1,000,000	1,000,000
EBT	6,000,000	7,200,000	8,000,000	7,000,000	7,400,000
Less: Tax (50%)	3,000,000	3,600,000	4,000,000	3,500,000	3,700,000
Net income	3,000,000	3,600,000	4,000,000	3,500,000	3,700,000
Shares outstanding	1,000,000	1,000,000	1,200,000	1,000,000	1,040,000
EPS (Rs)	**3.00**	**3.60**	**3.33**	**3.50**	**3.56**

Comment:

- Both issues raise EPS above the current Rs 3.00 because the new Rs 10 million earns 20% before tax while costing only 8% or 10%.
- **Convertible:** EPS is highest before conversion (Rs 3.60, lowest coupon); on conversion EPS falls to Rs 3.33 because 200,000 new shares are issued (dilution) although interest is saved.
- **Warrants:** EPS rises from Rs 3.50 to Rs 3.56 on exercise because exercise brings in Rs 2 million of new cash (which earns 20%) while the bonds remain, and only 40,000 shares are added.
- Fully diluted, the warrant alternative (Rs 3.56) gives a higher EPS than the convertible (Rs 3.33).

> 💡 **Exam Hint (remember):** Conversion ratio = Face value ÷ Conversion price; CS ↑ = new shares × par, APIC ↑ = new shares × (price − par); EPS = (EBIT − I)(1 − T) ÷ shares. Trap: on conversion, interest disappears and total capitalization stays the same; on warrant exercise, the bond and its interest remain and capitalization increases by the cash received. Shortcut: Convertible → debt moves to equity (no new cash); Warrant → new cash comes in (debt stays).

(iv) Reasons for issuing warrants and convertibles

A. Reasons for issuing convertibles

1. **Sweetener / lower interest cost:** The conversion option lets the firm issue debt at a lower coupon (8% versus 10% for straight bonds) and with fewer restrictive covenants.
2. **Deferred equity financing ("equity through the back door"):** The firm effectively sells common stock at a future date, at a conversion price (Rs 50) higher than the current market price (Rs 40), so fewer shares are issued than in a direct equity issue now — less dilution.
3. **Delayed dilution of EPS and control:** Dilution occurs only when the project starts generating earnings and conversion takes place.
4. **Forcing conversion with the call feature:** Once the share price exceeds the conversion price, the firm can call the bonds to force conversion, eliminating debt and improving the capital structure.
5. **Suitable for growing firms and low-rated firms:** Rapidly growing companies like HHC with uncertain risk can raise funds on acceptable terms; investors get downside protection of a bond and upside of equity.
6. **Reduced agency conflicts:** Convertibles reduce the bondholder–shareholder conflict over risky investments.

B. Reasons for issuing warrants

1. **Sweetener to debt:** Warrants make bonds more attractive, so the firm can issue them at lower interest, with fewer covenants, or when it otherwise could not borrow (small or risky firms).
2. **Additional cash inflow:** On exercise, the firm receives fresh capital (Rs 2 million for HHC) while the debt remains outstanding — useful for a growing firm that needs more funds as it grows.
3. **Sale of stock at a higher price:** Exercise price (Rs 50) is set above the current price (Rs 40), so equity is raised at a premium.
4. **Delayed dilution:** Warrants are exercised only when the share price rises above the exercise price — usually when the firm is performing well.
5. **Detachability and marketability:** Detachable warrants can be traded separately, increasing the attractiveness of the issue.
6. **Used in mergers, reorganizations and venture financing** as compensation or inducement to underwriters, promoters and investors.

Conclusion for HHC: Both instruments let HHC borrow cheaply now and raise equity later at a premium price. The convertible has a lower interest cost and eliminates the debt on conversion (lower leverage), while the warrant alternative brings in additional cash and gives higher fully diluted EPS (Rs 3.56 vs Rs 3.33) but keeps the debt and higher interest burden. The choice depends on whether HHC wants to reduce leverage (convertible) or needs more total funds and higher EPS (warrants).

Paper 2 — BNK 209: Insurance and Risk Management

Tribhuvan University · Faculty of Management · BBA 8th Semester · Model Question · Full Marks 100 · Time 3 Hrs

Group A — Brief Answer Questions (10 × 2 = 20)

Q1. Define risk, peril and hazard.

Solution:

- **Risk:** Uncertainty about the occurrence of a loss. In other words, it is the chance that the actual outcome will differ from the expected outcome (objective risk) or a person's doubt about the outcome (subjective risk).
- **Peril:** The *cause* of loss. Examples: fire, flood, earthquake, theft, accident, death.
- **Hazard:** A condition that *increases the frequency or severity* of loss. It has four types:
- *Physical hazard:* a physical condition such as icy roads, faulty wiring or a defective lock.
- *Moral hazard:* dishonesty or bad character that increases loss, e.g. faking an accident or inflating a claim.
- *Attitudinal (morale) hazard:* carelessness or indifference to loss because insurance exists, e.g. leaving the car unlocked.
- *Legal hazard:* features of the legal system or regulatory environment that increase loss, e.g. very large court awards.

Q2. What do you mean by systemic risk?

Solution:

Systemic risk is the risk that the failure of one institution or market segment spreads through interconnections and causes the collapse or serious disruption of the *entire* financial system or economy. Examples are the 2008 global financial crisis (failure of Lehman Brothers) and a bank run that spreads to many banks.

Key points:

- It arises from interconnectedness and contagion ("domino effect"). It is not caused by an event that hits everyone at once.
- It is different from **systematic risk**, which is economy-wide, non-diversifiable *market* risk such as inflation, interest-rate or recession risk.
- It cannot be handled by private insurance because losses are catastrophic and correlated. It is managed through regulation, capital and solvency requirements, central bank or lender-of-last-resort support, and supervision. In Nepal this is done by Nepal Rastra Bank and the Nepal Insurance Authority.

Q3. List out the objectives of risk management.

Solution:

Risk management has objectives before a loss happens and after it happens.

Pre-loss objectives:

1. **Economy:** prepare for potential losses in the most economical way (cost of safety programmes, premiums, etc.).
2. **Reduction of anxiety:** reduce worry and fear about possible losses.
3. **Meeting legal (externally imposed) obligations:** e.g. safety rules, compulsory third-party motor insurance, Social Security Fund contributions.

Post-loss objectives:

1. **Survival** of the firm.
2. **Continued operation** of the business.
3. **Stability of earnings.**
4. **Continued growth.**
5. **Social responsibility:** minimise the effect of the loss on employees, customers, suppliers and the community.

Q4. Define risk pooling.

Solution:

Risk pooling (pooling of losses) means spreading the losses of a few people over the whole group. Each member pays a small certain amount (the premium) in place of a large uncertain loss. In this way, the average loss replaces the actual loss.

Key points:

- It is the essence of insurance.
- It is based on the **law of large numbers**. The larger the pool of independent exposures, the more accurately actual losses can be predicted, and the lower the relative risk (variation) for each member.
- It requires many similar, independent (uncorrelated) exposure units.
- Example: if 1,000 houses each pay Rs 2,000 into a pool, the Rs 20,00,000 collected can pay for the few houses that actually burn.

Q5. Define insurance and list down its basic characteristics.

Solution:

Insurance is the pooling of fortuitous losses by transferring such risks to an insurer. In return for a premium, the insurer agrees to indemnify insureds for such losses, provide other pecuniary benefits when a loss occurs, or render services connected with the risk (Rejda).

Basic characteristics:

1. **Pooling of losses:** the losses of the few are spread over the many, and average loss replaces actual loss.
2. **Payment of fortuitous losses:** only losses that are unforeseen, unexpected and accidental are paid. Intentional losses are not paid.
3. **Risk transfer:** pure risk is transferred from the insured to an insurer who is financially stronger.
4. **Indemnification:** the insured is restored, partly or fully, to the financial position held before the loss.

Q6. Differentiate life insurance and non-life insurance.

Solution:

Basis	Life Insurance	Non-Life (General) Insurance
Subject matter	Human life (death, survival to maturity)	Property, liability, motor, health, agriculture, etc.
Nature of contract	Contract of assurance / benefit (valued)	Contract of indemnity
Event	Death is certain; only its timing is uncertain	Loss may or may not occur
Duration	Long-term (10–40 years or whole life)	Usually short-term (typically 1 year, renewable)
Insurable interest	Required only at the time the contract starts	Required at the time of loss (and usually at start)
Savings/investment element	Often present (endowment, whole life)	Absent; pure protection
Subrogation & contribution	Not applicable	Applicable
Surrender value	Available in many plans	Not available
Examples in Nepal	Rastriya Beema Sansthan, Nepal Life, LIC Nepal	Rastriya Beema Company, Shikhar, Sagarmatha Lumbini, etc.

Q7. Define loss forecasting. List out the common techniques of loss forecasting.

Solution:

Loss forecasting means predicting the likely frequency and severity of future losses from past loss data and relevant factors. Its purpose is to choose suitable risk-control and risk-financing techniques, set retention levels and budget for losses.

Common techniques:

1. **Probability analysis:** assigns probabilities to events based on past data, e.g. the chance of fire in a year.
2. **Regression analysis:** relates losses to one or more variables such as payroll, number of employees, output or sales. For example, workers' compensation claims are predicted from the number of employees.
3. **Forecasting based on loss distributions:** fits past losses to a statistical distribution, such as the normal distribution, to estimate the expected loss and the range of loss.
4. **Trend analysis:** projects past losses into the future after adjusting for inflation and growth.

Q8. How does Risk Management Information System help in risk management?

Solution:

A **Risk Management Information System (RMIS)** is a computerised database that stores and analyses risk management data. It holds property values, loss history, claims, policies, premiums and exposure data.

How it helps:

- It provides a **property schedule** and an **exposure database**, which help in identifying loss exposures.
- It keeps track of **losses, claims and loss reserves**. This makes loss forecasting and trend analysis easier.
- It helps in **comparing insurance options**, renewing policies and allocating insurance costs among departments.
- It supports **monitoring and reporting** to management and to regulators. In this way it improves decision-making and keeps the programme cost-effective.

Q9. What are the tax benefits and payout taxes for life insurance in Nepal?

Solution:

As per current Income Tax Act, 2058 provisions (verify against the latest Finance Act):

Tax benefits (on premium):

- **Life insurance premium** paid by a resident individual is deductible from taxable income. The limit is the actual premium or **Rs 40,000 per year**, whichever is lower.
- **Health insurance premium** is deductible up to **Rs 20,000 per year**.
- Contributions to approved retirement funds (EPF, CIT, SSF) are also deductible within the prescribed limits.

Payout taxes:

- The **sum assured received on death** of the insured, received by the nominee, is generally **tax-exempt**.
- The **maturity benefit** of a life policy is generally exempt for the insured. However, the investment-type gain or return component may attract **TDS of about 5%** where applicable.
- Surrender of a policy before the prescribed period may lead to the tax-deduction benefit being withdrawn or taxed.

Q10. Define third-party liability insurance in terms of auto (motor) insurance.

Solution:

Third-party liability motor insurance covers the vehicle owner's or driver's **legal liability to other persons (the third party)** for **death, bodily injury or property damage** caused by the use of the insured vehicle.

Key points:

- There are three parties: the insured (first party), the insurer (second party) and the injured outsider (third party).
- It is **compulsory** in Nepal under the **Motor Vehicle and Transport Management Act, 2049** and the directives of the Nepal Insurance Authority. A vehicle cannot be renewed or registered without it.

- It does **not** cover **own damage** to the insured vehicle. Own damage needs a comprehensive policy.
- The premium is fixed by the regulator according to the type and capacity of the vehicle.

Group B — Descriptive Answer Questions (Any six, 6 × 5 = 30)

Q11. Differentiate between insurance and gambling.

Solution:

Insurance and gambling look alike because in both, a small amount is paid and a large amount may be received on an uncertain event. However, they are fundamentally different.

Basis	Insurance	Gambling
Nature of risk	Deals with an **existing pure risk** (loss or no loss)	**Creates a new speculative risk** (loss, gain or no change)
Purpose	Protection against loss	Profit or gain
Social value	**Socially productive:** both insurer and insured gain when no loss occurs	**Socially unproductive:** the winner gains only at the loser's expense
Indemnity	Restores the insured to the pre-loss position; no profit allowed	No indemnity; the gambler can gain far more than the stake
Insurable interest	Must be present; the insured must suffer a financial loss	Not required
Basis of prediction	Based on the **law of large numbers** and scientific premium calculation	Based on chance; no pooling of losses
Effect on society	Reduces uncertainty and promotes savings, investment and credit	Increases uncertainty; may cause addiction and social problems
Legal status	Legal, regulated contract (by NIA in Nepal)	Often illegal or heavily restricted

Conclusion: Insurance is a risk-*reducing* device that transfers and pools an existing risk. Gambling is a risk-*creating* activity. Therefore insurance is a legitimate financial service and gambling is not.

Q12. Explain the concept and significance of law of large number in risk pooling?

Solution:

Concept

The **law of large numbers** says that as the number of exposure units (the sample) increases, the actual results get closer to the expected results. Put another way, the relative variation of actual losses from expected losses becomes smaller.

For example, if the probability of a house fire is 1%, an insurer with 10 houses cannot predict losses. An insurer with 100,000 similar houses can expect about 1,000 fires and will be fairly accurate.

Conditions for it to work

- There must be a large number of **homogeneous**, similar exposure units.
- Losses must be **independent** (not correlated or catastrophic).
- The loss must be **accidental** and **measurable**.

Significance in risk pooling

1. **Accurate prediction of losses:** the insurer can estimate the expected loss and charge a fair premium.
2. **Reduction of objective risk:** risk falls in proportion to 1/√N. As the pool grows, uncertainty for the insurer falls.
3. **Basis for premium calculation:** the expected loss per unit becomes the pure premium, and loading is added to it.
4. **Makes insurance affordable:** each member pays a small, certain premium in place of a large, uncertain loss.
5. **Financial stability of insurers:** predictable claims allow proper reserves and solvency.
6. **Explains why insurers avoid catastrophic risks:** correlated losses break the law. Examples are earthquake losses in one area, as in Nepal in 2015. Such risks need reinsurance, for example with Nepal Re-Insurance Company.

Conclusion: The law of large numbers is the mathematical foundation of insurance. It turns individual uncertainty into group predictability.

> **Q13. Define adverse selection. Explain the tools used by insurance companies to control the chance of adverse selection.**

Solution:

Adverse selection is the tendency of people with a **higher-than-average chance of loss** to seek insurance at **standard (average) rates**. If this is not controlled, actual losses become higher than expected. It arises from **asymmetric information**: the applicant knows more about their own risk than the insurer does. Examples are a person with heart disease buying life or health insurance, or a house owner in a flood zone buying flood cover.

Tools to control adverse selection

1. **Careful underwriting:** the insurer selects and classifies applicants. It may accept them at standard rates, accept them at higher rates, or reject them.
2. **Medical examination and proposal form:** medical tests, health questionnaires and the duty of utmost good faith reveal hidden risk factors.
3. **Risk classification and rating:** applicants are grouped by age, health, occupation and location. Higher-risk groups pay higher premiums, for example substandard rating with extra premium.
4. **Policy provisions:** - *Pre-existing condition exclusion* and *waiting period* in health insurance. - *Suicide clause:* suicide within the first 1–2 years is not covered. - *Incontestable or contestable period:* the insurer can contest the policy for misrepresentation within 2 years. - Exclusions for war, hazardous activities and similar risks.
5. **Deductibles and coinsurance:** the insured shares part of the loss. This discourages high-risk or careless buyers.

6. **Information sharing:** insurers share claim data, the motor claim history and blacklists through the regulator or industry databases.
7. **Group underwriting:** people are insured as a group (for example, employees) who joined for reasons other than buying insurance. This dilutes individual adverse selection.

> **Q14. Explain about premature death. Also explain its financial implications.**

Solution:

Premature death is the death of a family head with **unfulfilled financial obligations**. These include dependants to support, children to educate, and a mortgage or other debts to pay off. A death is "premature" because of these financial responsibilities, not only because of age.

Financial implications (costs of premature death)

1. **Loss of human life value:** the family permanently loses its share of the deceased's future earnings. This loss can be very large.
2. **Additional expenses:** funeral expenses, medical bills of the last illness, estate settlement costs and unpaid debts.
3. **Insufficient income for the family:** many families do not have enough savings or insurance. Their living standard falls, and children's education may stop.
4. **Non-economic costs:** emotional grief, loss of a role model, loss of guidance for children, and loss of household services.

When premature death causes financial insecurity

- When the deceased has **dependants** (spouse, children, parents).
- When there are **outstanding debts** such as a home loan or car loan.
- When there are **children to educate**.
- When a **single-parent** family loses its only earner.
- When there is **insufficient life insurance** or savings.

A person with no dependants, for example a single person, causes little financial hardship to others.

Management: Life insurance (term or whole life), social security benefits (dependent family benefits under the SSF in Nepal), and savings, EPF and CIT balances reduce these financial consequences.

> **Q15. Mr. Suman Thapa, aged 40, is married and has two children. He earns Rs.600,000 annually and plans to retire at age 60. Of this amount, Rs.30,000 is used for taxes, life and health insurance, and his personal needs. The remaining amount is used to support his family. Using a discount rate of 6 percent, calculate the human life value of Mr. Shrestha. Note that the present value of Re.1 received annually for 20 years at 6% is Rs.11.4699.**

Solution:

Note: The question names "Mr. Suman Thapa" first and "Mr. Shrestha" later. Both are taken to be the same person.

Given:

- Present age = 40 years; retirement age = 60 years
- Annual earnings = Rs 600,000
- Self-maintenance costs (taxes, life & health insurance, personal needs) = Rs 30,000 per year
- Discount rate (r) = 6%
- PVIFA (6%, 20 years) = 11.4699

Formula:

Human Life Value (HLV) = Annual amount available to family × PVIFA(r, n)

where annual amount available to family = Annual earnings − Self-maintenance costs, and n = Retirement age − Present age.

Step-by-step calculation:

Step 1: Working years remaining n = 60 − 40 = **20 years**

Step 2: Annual amount used to support the family = 600,000 − 30,000 = **Rs 570,000**

Step 3: Present value of this annual amount for 20 years at 6% HLV = 570,000 × 11.4699 = (570,000 × 11) + (570,000 × 0.4699) = 6,270,000 + 267,843 = **Rs 6,537,843**

Answer: The human life value of Mr. Suman Thapa (Mr. Shrestha) is **Rs 6,537,843** (about Rs 65.38 lakh). This is the amount of life insurance needed to replace his economic value to his family.

> 💡 **Exam Hint (remember):** Formula: **HLV = (Annual earnings − Self-maintenance expenses) × PVIFA(r, n)**, where n = retirement age − current age. Common trap: forgetting to **deduct personal expenses, taxes and insurance**. Using Rs 600,000 gives the wrong answer of Rs 6,881,940. Another trap is using retirement age (60) as n instead of the years left (20). Shortcut: "Family share × annuity factor": 5.7 lakh × 11.4699 ≈ 65.38 lakh.

Q16. Explain the third-party liability insurance in terms of auto (motor) insurance.

Solution:

Meaning

Third-party liability motor insurance protects the owner or driver of a vehicle against the **legal liability** to pay compensation to a **third party** (any person other than the insured and the insurer). The liability must arise from **death, bodily injury or property damage** caused by an accident involving the insured vehicle. The insurer pays the third party on behalf of the insured.

Parties involved

- *First party:* the insured vehicle owner.
- *Second party:* the insurance company.
- *Third party:* the victim, such as a pedestrian, a passenger of another vehicle, or the owner of damaged property.

Coverage

1. **Death or bodily injury** of third parties, including medical expenses, up to the prescribed limits.
2. **Damage to third-party property**, such as another vehicle, a house or a wall, up to the prescribed limit.
3. **Legal costs** of defending claims, as provided in the policy.
4. In Nepal, the tariff usually also includes cover for the **driver, helper/conductor and passengers**.

Exclusions

- **Own damage** to the insured vehicle (needs a comprehensive policy).
- Driving without a valid licence, under the influence of alcohol or drugs, or for an unauthorised purpose.
- Intentional damage, war, and nuclear risks.
- Contractual liability.

Legal position in Nepal

- It is **compulsory** under the **Motor Vehicle and Transport Management Act, 2049**. A vehicle cannot legally run on the road, nor be renewed (bluebook), without valid third-party insurance.
- The **Nepal Insurance Authority** fixes the uniform premium tariff and the compensation limits by vehicle type and capacity.

Importance

- It ensures that **accident victims get compensation** even if the driver cannot pay.
- It protects vehicle owners from large, unexpected legal liabilities.
- It is cheap compared with a comprehensive policy, and it promotes social justice and road safety.

Q17. Explain social insurance with few examples.

Solution:

Meaning

Social insurance is a government insurance programme with certain characteristics that set it apart from other government programmes. It provides protection against social risks such as old age, disability, death, sickness, unemployment and work injury. It is usually **compulsory** and **financed by contributions** from employers, employees and sometimes the government.

Characteristics

1. **Compulsory programmes:** coverage is mandatory for eligible workers.
2. **Floor of income:** it provides only a minimum, basic level of protection. Individuals add private insurance on top.
3. **Emphasis on social adequacy:** benefits favour low-income groups. This means **income redistribution**, rather than strict individual equity.
4. **Benefits loosely related to earnings:** higher contributions usually give somewhat higher benefits.
5. **Benefits prescribed by law:** they are not set by an individual contract.
6. **No needs (means) test:** benefits are paid as a right if eligibility conditions are met.

7. **Full funding not necessary:** benefits are often financed on a pay-as-you-go basis.
8. **Financed mainly by contributions:** from covered employees and employers (for example, payroll taxes).

Examples – Nepal

- **Social Security Fund (SSF):** a contribution-based scheme under the Contribution-based Social Security Act, 2075. It includes the medical, health and maternity scheme; the accident and disability scheme; the dependent family scheme; and the old-age protection scheme. The total contribution is 31% (11% employee + 20% employer) of basic salary.
- **Employees Provident Fund (EPF):** compulsory retirement savings for government and other employees.
- **Citizen Investment Trust (CIT):** retirement and gratuity funds.
- **Health Insurance Board scheme:** national health insurance.

Examples – International (USA)

- **OASDI (Old-Age, Survivors and Disability Insurance):** the Social Security programme.
- **Unemployment insurance.**
- **Workers' compensation** (for job-related injury or disease).
- **Medicare** (health insurance for the elderly).

Group C — Analytical Answer Questions (Any three, 3 × 10 = 30)

Q18. Explain the concept and process of risk management.

Solution:

Concept of risk management

Risk management is a **systematic process for identifying and evaluating pure loss exposures** faced by an organisation or individual, and for **selecting and implementing the most appropriate techniques** for treating those exposures (Rejda). Its aim is to protect assets, earnings and people at the lowest possible cost. The broader **Enterprise Risk Management (ERM)** approach covers pure, speculative, strategic, financial and operational risks together.

It has pre-loss objectives (economy, reduction of anxiety, meeting legal obligations) and post-loss objectives (survival, continued operation, stable earnings, continued growth and social responsibility).

Process of risk management

Step 1: Identify loss exposures

All potential losses are identified:

- *Property exposures:* buildings, equipment, inventory, vehicles.
- *Liability exposures:* defective products, environmental pollution, professional errors.
- *Business income exposures:* loss of profit due to interruption.

- *Human resource exposures:* death, disability or retirement of key employees, and work injury.
- *Crime, cyber, and foreign/political exposures.*

Sources of information include questionnaires, physical inspection, flowcharts, financial statements, past loss data and the RMIS.

Step 2: Measure and analyse the loss exposures

Each exposure is estimated for:

- **Loss frequency:** the probable number of losses.
- **Loss severity:** the probable size of losses, including the *maximum possible loss* and the *probable maximum loss*.

Exposures are then ranked by importance. Severity is more important than frequency because a single catastrophic loss can destroy a firm.

Step 3: Select the appropriate combination of techniques

A. Risk control (reduce frequency or severity):

- **Avoidance:** not acquiring or abandoning the exposure, e.g. not producing a risky product.
- **Loss prevention:** reducing *frequency*, e.g. safety training, maintenance.
- **Loss reduction:** reducing *severity*, e.g. sprinklers, fire alarms, rehabilitation.
- **Duplication:** keeping back-ups or spares, e.g. data backups.
- **Separation:** dividing assets among several locations.
- **Diversification:** spreading business across products, customers or markets.

B. Risk financing (pay for losses that occur):

- **Retention:** the firm pays losses itself, through current income, funded reserves, a credit line or a captive insurer. It is suitable for high-frequency, low-severity losses.
- **Non-insurance transfers:** contracts, leases, hold-harmless agreements, hedging, incorporation.
- **Commercial insurance:** suitable for low-frequency, high-severity losses. It involves choosing the insurer, coverage, deductibles and limits.

Risk management matrix:

	Low severity	**High severity**
Low frequency	Retention	Insurance
High frequency	Loss prevention + retention	Avoidance / loss reduction

Step 4: Implement and monitor the risk management programme

- Prepare a **risk management policy statement**.
- Coordinate with other departments (accounting, finance, HR, production).
- **Monitor and review** the programme periodically, because exposures, laws and costs change. Use risk management audits and the RMIS.

Conclusion: Risk management is a continuous cycle: identify, measure, treat and monitor. It lets an organisation lower the cost of risk, maintain stability and achieve its goals.

Q19. Explain the fundamental legal principles of insurance contract.

Solution:

An insurance contract is a legal contract. It must meet general contract requirements and also follow certain special principles.

A. Fundamental legal principles

1. Principle of indemnity

The insurer should restore the insured to approximately the **same financial position** held before the loss. The insured should not profit. This prevents overinsurance and reduces moral hazard. Loss is measured by:

- **Actual cash value (ACV)** = replacement cost − depreciation (or fair market value, or the broad evidence rule).
- **Replacement cost** insurance: no deduction for depreciation.
- **Valued policy:** a fixed amount is paid for total loss, e.g. antiques or paintings.

Exceptions: valued policies, valued policy laws, replacement cost insurance, and **life insurance** (a valued contract, because human life cannot be measured).

2. Principle of insurable interest

The insured must stand to **lose financially** if the loss occurs. Its purposes are to prevent gambling, reduce moral hazard and measure the loss. In property insurance it must exist **at the time of loss**. In life insurance it must exist **only at the inception** of the policy. Self, spouse, dependants, creditors and business partners have insurable interest.

3. Principle of subrogation

After paying the loss, the insurer takes over the insured's **legal rights to recover** from a negligent third party. Its purposes are to prevent the insured from collecting twice, hold the guilty party responsible, and keep premiums low. It does not apply to life insurance or most personal health insurance.

4. Principle of utmost good faith (uberrimae fidei)

Both parties must be completely honest. This principle is supported by three doctrines:

- **Representations:** statements made by the applicant. A *material, false* representation that is *relied upon* makes the contract voidable.
- **Concealment:** intentional failure to disclose a material fact makes the contract voidable.
- **Warranty:** a statement or promise that is part of the contract, such as "a watchman will be on duty". A breach allows the insurer to deny the claim.

5. Principle of proximate cause

The insurer is liable only if the **dominant, effective cause** (causa proxima) of the loss is an insured peril. Example: fire caused by lightning is covered, while fire caused by war, if war is excluded, is not.

6. Principle of contribution

When the same property is insured with **more than one insurer**, the insurers share the loss in proportion to their coverage (pro-rata liability). The insured cannot recover more than the actual loss. This is a corollary of indemnity.

7. Principle of loss minimisation

The insured must take reasonable steps to minimise the loss, acting as an uninsured prudent person would.

B. Requirements of a valid insurance contract

1. **Offer and acceptance:** usually, the proposal by the applicant is the offer, and the issue of the policy by the insurer is the acceptance.
2. **Consideration:** the premium and the promise to follow policy conditions.
3. **Competent parties:** the parties must be of legal age and sound mind, and the insurer must be licensed (by the NIA in Nepal).
4. **Legal purpose:** contracts that encourage illegal acts are void.

C. Distinct legal characteristics

Characteristic	Meaning
Aleatory	Values exchanged are unequal and depend on chance
Unilateral	Only the insurer makes a legally enforceable promise
Conditional	The insurer must pay only if the insured meets the policy conditions
Personal	Property insurance insures the person, not the property; assignment needs consent
Contract of adhesion	The insurer drafts the terms; ambiguities are interpreted in favour of the insured

Conclusion: These principles keep insurance fair, prevent gambling and fraud, and preserve its basic purpose of indemnity.

Q20. Explain about agriculture insurance. Why do you think agriculture insurance is important for countries like Nepal?

Solution:

Concept of agriculture insurance

Agriculture insurance protects farmers against **financial losses to crops, livestock, poultry and fishery** caused by natural perils, diseases and other uncontrollable events. It compensates them so they can continue farming.

Types of agriculture insurance

1. **Crop insurance:** covers paddy, maize, wheat, vegetables, fruits, tea, coffee, etc. against drought, flood, hailstorm, landslide, pest and disease.
2. **Livestock insurance:** covers cattle, buffalo, goats, pigs, etc. against death due to disease or accident.
3. **Poultry insurance:** covers birds against death due to disease (e.g. bird flu) or accident.
4. **Fishery insurance:** covers fish ponds against disease, flood and pollution.
5. **Others:** bee-keeping, greenhouse/tunnel farming and plantation insurance.

Approaches

- **Indemnity-based insurance:** compensation is based on the actual assessed loss of the farm. It is accurate but has high assessment costs and moral hazard.
- **Index-based (parametric) insurance:** payment is triggered by an index such as rainfall, temperature or area yield. There is no farm-level loss assessment, settlement is

fast and costs are low, but there is *basis risk* (the farm loss may not match the index).

Agriculture insurance in Nepal

- It started under the Agriculture Insurance Directive (2069 B.S.) issued by Beema Samiti, now the **Nepal Insurance Authority (NIA)**.
- The government provides a **premium subsidy of approximately 80%** (as per the directive). The farmer pays only about 20%.
- Non-life insurers are mandated to provide agriculture and livestock insurance. Reinsurance support comes from **Nepal Re-Insurance Company**.

Importance of agriculture insurance for Nepal

1. **Dependence on agriculture:** over 60% of the population depends on agriculture for a living, and it contributes about a quarter of GDP. Protecting farmers protects the economy.
2. **High exposure to climate and natural risks:** monsoon floods, landslides, drought, hailstorms, cold waves and climate change cause frequent and severe crop and livestock losses.
3. **Income stability and poverty reduction:** compensation prevents farmers from falling into debt traps and distress sales of assets.
4. **Access to credit:** insured crops and livestock act as collateral. Banks and cooperatives lend more readily, which supports agricultural financing.
5. **Encourages commercialisation and modern technology:** farmers can take the risk of investing in high-value crops, improved breeds and commercial farms.
6. **Food security:** quick recovery after losses maintains production and national food supply.
7. **Reduces the government's disaster relief burden:** a planned risk transfer replaces ad hoc relief.
8. **Reduces migration:** more stable farm income reduces youth migration abroad.
9. **Rural development and financial inclusion:** brings rural farmers into the formal financial system.

Challenges in Nepal: low awareness, small and fragmented landholdings, difficult loss assessment, moral hazard and fraud, delays in claim settlement, limited data for index insurance, and low insurer interest because of high costs.

Way forward: awareness programmes, index-based products, digital claim processes, timely release of subsidy and cooperative-based distribution.

Conclusion: For an agrarian, disaster-prone country like Nepal, agriculture insurance is a vital tool for protecting farmers' livelihoods, ensuring food security and promoting agricultural growth.

> **Q21. Insurance is one of the heavily regulated area of the overall financial system. In light of this statement, explain the reasons for insurance regulations.**

Solution:

Introduction

Insurance is regulated heavily because insurers collect premiums in advance and promise to pay large sums far into the future. Contracts are complex, and the public depends on

insurers to remain solvent. Failure of an insurer harms policyholders and the economy. In Nepal, insurance is regulated by the **Nepal Insurance Authority (NIA)**, formerly Beema Samiti, under the **Insurance Act, 2079**.

Reasons for insurance regulation

1. Maintain insurer solvency

This is the most important reason. Premiums are paid in advance, but claims may be paid many years later, especially in life insurance. Regulation ensures that insurers have enough capital, reserves and solvency margins, and sound investments, so they can pay claims. Insolvency would cause great hardship to policyholders.

2. Compensate for inadequate consumer knowledge

Insurance contracts are technical and complex. Most consumers cannot judge policy terms, prices or the financial strength of insurers. Regulators approve policy forms and provide information to protect consumers.

3. Ensure reasonable rates

Rates should be **adequate** (to keep insurers solvent), **not excessive** (to avoid overcharging) and **not unfairly discriminatory**. In Nepal, some rates are tariff-based, e.g. motor third-party and agriculture.

4. Make insurance available

Regulation ensures availability of essential coverages to all, including high-risk or underserved segments. Examples are compulsory third-party motor cover, subsidised agriculture and livestock insurance, micro-insurance, and the requirement to open branches in rural areas.

5. Prevent unfair practices and protect policyholders

Regulation prevents misleading advertising, twisting, rebating, misrepresentation by agents and unfair claim practices. It also handles consumer complaints and ensures fair and prompt claim settlement.

6. Insurance is affected with a public interest

Insurance supports credit, trade and families' financial security, and it mobilises long-term savings for investment. Its failure has wide social and economic consequences, and it is a source of potential systemic risk.

Methods of regulation

- **Legislation:** acts passed by parliament, e.g. the Insurance Act, 2079.
- **Courts:** decisions interpreting contracts and the constitutionality of laws.
- **Regulatory authority:** a government agency, the **NIA**, issues regulations, directives and circulars, licenses insurers and agents, and supervises them.

Areas regulated

Area	What is regulated
Formation & licensing	Minimum paid-up capital, licence of insurers, brokers, agents and surveyors
Solvency	Capital adequacy, solvency margin, reserves, investment directives, reinsurance (e.g. cession to Nepal Re)
Rates	Rate approval, tariffs

Area	What is regulated
Policy forms	Approval of new products and policy wording
Sales practices & consumer protection	Agent licensing and conduct, commission limits, grievance handling
Claims	Timely settlement, surveyor appointment
Financial reporting	Audited statements, actuarial valuation, NFRS compliance
Taxation	Premium tax, income tax on insurers
Liquidation	Rehabilitation and winding-up of weak insurers

Conclusion: Insurance is heavily regulated because it is a long-term promise of future payment and is affected with a strong public interest. Regulation protects policyholders, keeps insurers solvent and maintains public confidence in the financial system.

Group D — Comprehensive / Case Study (4 × 5 = 20)

Q22. Read the following case carefully and answer the question that follow:

Aarav Sharma is a 42 years old Senior Project Manager and the primary breadwinner for his family with a spouse and two children. He is also a stock investor and trader. Recently, he decided to quit his job and start his own venture named Everest Tech Solutions, a consultancy firm that specializes in digital transformation for local financial institutions. He has invested a significant portion of his personal savings into the startup, leased a modern office space, and hired five employees. As Everest Tech Solutions begins its operations, Aarav faces a complex web of risks that threaten both his personal financial stability and the viability of his new business.

Aarav's family relies entirely on his income. If he were to suffer a premature death, become permanently disabled or get diagnosed with a critical illness, his family would be unable to meet their living expenses and education costs for their children. Further, his family will not have funds to pay off the mortgage on their house and the car. The new office contains expensive server equipment and high-end laptops. A fire or a localized earthquake could result in a total loss of physical assets. There is also the risk of data breach on the client's server due to a configuration error his employee. The loss of sensitive customer data and business interruption could expose his business to enormous liability.

Aarav is planning to purchase a life insurance policy to ensures that in the event of his untimely demise, his family receives a lumpsum to clear the mortgage and fund his children's education. He is also worried that if he becomes permanently disabled due to accident or gets diagnosed with any critical illness, he will not be able to work and his regular income would be lost.

With years of experience, Aarav also knows there are insurance coverages available to cover the office and its contents (servers, furniture, and electronics) against fire, lightning, and natural disasters like earthquakes or floods. He is also aware about the professional liability insurance that can protect his company against data breaches or other losses due to negligence, errors and omission of his employees during their professional services. He believes the Professional Liability Insurance Policy will provide coverage for the damage to

the client as well as the legal costs to be paid for defending his organization. Aarav is also planning to take a group health insurance policy for his five employees to retain the employees as well as to ensure that they have access to quality healthcare without financial strain.

Case Study Questions

a) Identify all the pure risks and speculative risks faced by Aarav or his organization.

b) Which type of life insurance policy do you suggest to Aarav? Justify your answer. What additional benefits could be added to the policy to enhance the coverage?

c) How can a Professional Liability Insurance protect Everest Tech Solutions?

d) Why do you think is the decision to provide Group Health Insurance for the staffs considered a risk management tool, rather than just an additional expense to the organization?

Solution:

(a) Pure risks and speculative risks faced by Aarav / Everest Tech Solutions

A **pure risk** has only two possible outcomes: loss or no loss. It is generally insurable. A **speculative risk** has three possible outcomes: loss, gain or no change. It is generally not insurable.

Pure risks

Category	Pure risk identified in the case
Personal risks	Premature death of Aarav (family loses all income; mortgage, car loan and education unfunded)
	Permanent disability due to accident (loss of earning capacity)
	Critical illness (heavy medical cost + loss of income)
	Illness/injury of the five employees (medical costs, absenteeism)
	Loss of key person (Aarav himself, the business depends on him)
Property risks	Fire, lightning damaging servers, laptops, furniture, office contents
	Earthquake or flood causing total loss of physical assets
	Theft or burglary of equipment
	Business interruption – loss of income and extra expenses after a property loss
Liability risks	**Data breach** on the client's server due to an employee's configuration error
	Professional negligence, errors & omissions in consultancy services
	Loss of sensitive customer data → lawsuits, regulatory penalties
	Employer's liability for workplace injury of employees
	Liability arising from the leased premises and from his car (third-party motor liability)

Speculative risks

Speculative risk	Why speculative
Quitting a secure job to start Everest Tech Solutions	The venture may succeed (profit), fail (loss) or break even

Speculative risk	Why speculative
Investing personal savings in the startup	Return may be high, nil or negative
Stock investing and trading	Share prices can rise or fall
Leasing a modern office	The cost is fixed but the benefit depends on business growth
Hiring five employees	Payroll may or may not generate enough revenue
Market/competition risk in digital transformation for financial institutions	Demand, competition and technology changes
Taking a mortgage and car loan (financial leverage) and interest-rate changes	Asset value or rates may move favourably or unfavourably

Conclusion: Aarav can transfer the pure risks through life, health, property, business interruption and liability insurance. The speculative risks must be managed through business strategy, diversification and hedging.

(b) Recommended life insurance policy and riders

Suggested policy: Term life insurance, preferably a **level term plan up to age 60–65** with a high sum assured. Part of the cover may be **decreasing term (mortgage protection / reducing term)** to match the outstanding home and car loan balance.

Justification

1. **High need, limited cash:** Aarav is 42, his family depends entirely on him, and he has a mortgage, a car loan and two children's education to fund. He needs a **large lump sum**. Term insurance gives the **highest coverage at the lowest premium**.
2. **Business cash flow is tight:** most of his savings are invested in the startup. Expensive endowment or whole life premiums would strain liquidity. Under the "buy term and invest the difference" approach, savings can go into the business or investments.
3. **Temporary need:** the need is greatest *now*, during the next 15–20 years until the loan is repaid and the children are independent. Term cover matches this period exactly.
4. **Needs approach:** cover should equal the outstanding mortgage + car loan + children's education fund + income replacement for the family for several years + final expenses − existing assets. Only term insurance can affordably provide such a sum.
5. **Mortgage protection:** a decreasing term policy assigned to the bank ensures the house is not lost on his death.
6. **Later, once the business stabilises,** he may add a whole life or endowment policy for savings and estate purposes, and claim the tax deduction (up to Rs 40,000 premium).

Riders to enhance coverage

- **Accidental Death Benefit (ADB):** an extra sum assured if death is due to an accident.
- **Permanent Total/Partial Disability (PTD/PPD) benefit:** pays a lump sum or income if he is disabled by an accident. This directly addresses his worry about losing income.
- **Critical Illness rider:** a lump sum on diagnosis of cancer, heart attack, stroke, kidney failure, etc.
- **Waiver of Premium rider:** future premiums are waived if he becomes disabled or critically ill, so the policy continues.
- **Income/Family Benefit rider:** a monthly income to the family after his death, in addition to the lump sum.

- **Hospital cash / medical rider:** a daily allowance during hospitalisation.
- **Spouse or child term rider:** cover for other family members.

(c) How Professional Liability Insurance protects Everest Tech Solutions

Meaning: Professional Liability Insurance, also called **Errors & Omissions (E&O)** insurance, protects professionals and firms against **legal liability arising from negligence, errors, omissions or inadequate work** in providing professional services, which cause financial loss to clients. For an IT consultancy, it is often combined with **cyber liability** insurance.

How it protects the firm

1. **Covers damages to clients:** if an employee's configuration error causes a data breach or system failure at a client financial institution, the policy pays the **compensation, settlement or court judgment** awarded to the client.
2. **Legal defence costs:** it pays lawyers' fees, investigation and court costs, even if the claim is groundless. These costs can be huge for a startup.
3. **Data breach and cyber liability (if included or endorsed):** covers notification costs, forensic investigation, data restoration, regulatory fines (where insurable) and third-party claims for loss of sensitive customer data.
4. **Business interruption / network interruption losses** (under cyber extensions).
5. **Protects the firm's capital and survival:** a single large claim could bankrupt a new firm with limited capital and Aarav's personal savings. Transferring this low-frequency, high-severity risk is the correct risk-management response.
6. **Enhances credibility:** banks and financial institutions often *require* their IT vendors to carry E&O and cyber cover before awarding contracts. The policy therefore also helps win business.
7. **Covers employees' acts:** the firm is vicariously liable for its five employees' mistakes, and these are covered.

Key policy features to note

- Usually written on a **claims-made basis**: the claim must be made during the policy period.
- **Retroactive date:** errors committed after this date are covered.
- **Limits of liability** per claim and in aggregate, plus a **deductible**.
- **Exclusions:** intentional or fraudulent acts, bodily injury or property damage (covered under general liability), contractual penalties, and prior known claims.

Conclusion: With adequate limits, this policy transfers Everest Tech's most dangerous liability exposure to an insurer. It protects both the client relationship and the firm's survival.

(d) Group Health Insurance as a risk management tool rather than an expense

Group health insurance covers all five employees under one master policy for hospitalisation and medical expenses. It is a **risk management tool** because it reduces or transfers several loss exposures of the firm:

1. **Transfer of employee health risk:** the uncertain cost of employees' illness or injury (which the employer might otherwise feel obliged to pay) is converted into a small, certain, budgeted premium.

2. **Reduces human resource / key-person loss exposure:** in a five-person knowledge firm, each employee is critical. Health cover improves **retention** and reduces **turnover**, along with the costs of losing trained staff, recruiting, retraining and delaying client projects.
3. **Loss reduction:** employees seek **early treatment** without financial worry. This reduces the severity of illness and the number of days absent.
4. **Higher productivity and morale:** employees who feel financially secure are more focused and motivated. This also indirectly lowers the risk of errors that could cause professional liability claims.
5. **Business continuity:** healthy staff and quick recovery mean projects are delivered on time and business interruption is avoided.
6. **Cost efficiency through group underwriting:** group policies have **lower per-head premiums**, usually no individual medical examination, and cover for pre-existing conditions. Adverse selection is controlled because employees join for employment, not for insurance.
7. **Tax advantage:** the premium is a **deductible business expense**, so the net cost is lower.
8. **Legal and social compliance:** it helps meet the employer's obligations toward employee welfare under the Labour Act and Social Security Fund medical schemes. It also demonstrates social responsibility, which is a post-loss objective of risk management.
9. **Attracts talent:** it gives a startup a competitive advantage in hiring skilled IT professionals.

Conclusion: The premium is not a mere cost. It is an **investment that protects the firm's most valuable asset, its people**, and stabilises earnings. This is why group health insurance is a deliberate risk management decision for Everest Tech Solutions.

Paper 3 — BNK 202: Financial Derivatives

Tribhuvan University · Faculty of Management · BBA 8th Semester · Model Question 2026 · Full Marks 100 · Pass Marks 50 · Time 3 Hrs

Group A — Brief-answer Questions (10 × 2 = 20)

Q1. Why is a derivative instrument known as a zero-sum game?

Solution:

A derivative is called a zero-sum game because **whatever one party gains, the counterparty loses by exactly the same amount** (ignoring transaction costs).

- Every derivative contract has two sides — a long (buyer) and a short (seller/writer). It is a side-bet on the price of the underlying asset; no new wealth is created.
- Example: if a futures price rises by Rs 10, the long gains Rs 10 and the short loses Rs 10; the sum of gains and losses is zero.
- In options, the premium received by the writer equals the premium paid by the buyer, and the buyer's payoff at expiry is the writer's loss.
- After commissions and taxes it becomes slightly negative-sum for traders, but the derivative itself only transfers risk and wealth between parties.

Q2. Why do investors pay more for an American option than similar European option?

Solution:

Investors pay more for an American option because it gives **everything a European option gives plus the right of early exercise** at any time up to expiry.

- A European option can be exercised only on the expiration date; an American option can be exercised on or before that date.
- Extra rights never reduce value, so American option price ≥ European option price (same S, X, T, σ, r).
- Early exercise can be valuable — e.g., a deep in-the-money American put (receive X now and earn interest on it) or an American call just before a large dividend.
- The difference (early-exercise premium) is the price of this flexibility; for a call on a non-dividend-paying stock it is zero, so both are equal.

Q3. An investor who wants to achieve maximum gain and accept a higher maximum loss selects a call option with a lower exercise price. Justify.

Solution:

The statement is correct: a lower-strike call costs more (higher maximum loss) but pays more at every expiry price (higher gain).

- A call with a lower exercise price is more in-the-money, so its premium is higher. Since the maximum loss of a long call = premium paid, the **maximum loss is higher**.
- Profit at expiry = max(0, $S_T - X$) − C. For any S_T above the strikes, a lower X gives a larger payoff ($S_T - X$), and the extra payoff exceeds the extra premium when the price

rises strongly.

- The lower-strike call also starts paying off at a lower stock price, so the chance of gain is higher.
- Hence an aggressive investor who wants the larger possible gain and can bear a larger premium loss selects the lower exercise price (e.g., X = 100, C = 12 vs X = 110, C = 6: at S_T = 140 profit is 28 vs 24; max loss is 12 vs 6).

Q4. A longer-lived option has a higher price. Do you agree? Give reasons.

Solution:

Yes, generally a longer-lived option is worth more because it has **more time value**.

- More time means a greater chance that the stock price moves favourably (higher probability of ending deep in-the-money), while the loss is still limited to the premium.
- For **American options** (calls and puts) it is always true: a longer option has all the exercise opportunities of the shorter one plus more, so it can never be worth less.
- For **European calls** on non-dividend stocks it is also true, as longer time lowers the PV of the exercise price.
- Exception: a **European put** (deep in-the-money, where receiving X later lowers its PV) or a European call on a stock paying a large dividend before the longer expiry may be worth less.

Q5. Define future contract.

Solution:

A futures contract is a **standardised, exchange-traded agreement between two parties to buy or sell a specified quantity of an asset at a price fixed today (the futures price) for delivery on a specified future date**.

- **Standardised**: quantity, quality, delivery date and place are fixed by the exchange.
- **Clearinghouse**: it becomes buyer to every seller and seller to every buyer, eliminating default risk.
- **Margin and marking-to-market**: initial and maintenance margins are required and gains/losses are settled daily.
- **Liquidity/offsetting**: most positions are closed by an opposite trade before delivery. Unlike a forward, it is not customised or private.

Q6. What do you mean by upside capture in portfolio insurance?

Solution:

Upside capture is the **percentage of the uninsured portfolio's value (or gain) that the insured portfolio retains when the market rises**.

- Upside capture = Value of insured portfolio ÷ Value of uninsured portfolio (in a rising market) × 100.
- In portfolio insurance with protective puts, part of the funds is spent on put premiums, so fewer shares are held. When prices rise, the puts expire worthless and the insured portfolio is worth less than the uninsured one.
- Therefore upside capture is always less than 100%; the shortfall is the cost of insurance.

- Example: insured value Rs 92,300 vs uninsured Rs 100,000 → upside capture = 92.3%.

Q7. Under what conditions is a long hedge using futures suitable?

Solution:

A long hedge (buying futures) is suitable when a firm **plans to buy an asset in the future and fears that its price will rise** (i.e., it is effectively short in the spot market).

- Example: a flour mill that must buy wheat in three months, an airline that must buy jet fuel, or an importer who must pay in dollars later.
- Buying futures today locks in the purchase price: if the spot price rises, the gain on futures offsets the higher cost of buying the asset.
- It is also used by someone who has short-sold an asset and must buy it back later (anticipatory hedge).
- Suitable when futures on the same or a highly correlated asset are available, so basis risk is small.

Q8. Write the meaning of synthetic instruments.

Solution:

Synthetic instruments are **combinations of other securities/derivatives that replicate the payoff of a given instrument** exactly, so they should have the same price (law of one price).

- They are based mainly on put–call parity: S + P = C + X/(1 + r)^T.
- Synthetic call = long stock + long put − borrow PV(X); synthetic put = long call − long stock + lend PV(X).
- Synthetic stock = long call − long put + lend PV(X); synthetic forward/futures = long call + short put (same strike).
- Uses: to create a position when the actual one is unavailable or expensive, and to exploit arbitrage when actual and synthetic prices differ.

Q9. Why do investors prefer bull spreads from call options over those from put options?

Solution:

Both spreads profit from a moderate price rise, but a call bull spread (buy low-strike call, sell high-strike call) is often preferred because:

- **Net debit, known risk**: the investor pays a net premium upfront; the maximum loss equals this premium and is known in advance. A put bull spread (buy low-strike put, sell high-strike put) produces a net credit but requires the writer to post **margin**.
- **Less early-assignment risk**: the short high-strike put in a put spread is in-the-money when prices fall and can be exercised early (American puts); the short out-of-the-money call in a call spread is rarely exercised early.
- **Simpler and intuitive**: buying calls to gain from a rise is easier to understand and has lower margin/cash-management problems.
- The put version is chosen only when it offers a better net premium.

Q10. If you want to change the fixed-rate liability in to foliating-rate liability using a swap, how would you do it?

Solution:

Enter a plain-vanilla interest rate swap as the **fixed-rate receiver and floating-rate payer** (a pay-floating, receive-fixed swap).

- The firm continues to pay the fixed coupon on its existing loan/bond.
- Under the swap, it receives a fixed rate (which offsets the fixed coupon) and pays a floating rate (e.g., SOFR + spread).
- Net cost = Fixed paid on loan − Fixed received in swap + Floating paid ≈ Floating rate (± spread).
- Example: loan at 10% fixed; swap: receive 9.5% fixed, pay SOFR → net cost = SOFR + 0.5%, i.e., a floating-rate liability. Useful when rates are expected to fall.

Group B — Short-answer Questions (Any six, 6 × 5 = 30)

Q11. Describe the role of the derivative market.

Solution:

Derivative markets (options, forwards, futures and swaps) play an important role in modern financial systems. Their main roles are:

1. **Risk management (hedging):** The most important role. Derivatives allow firms and investors to transfer unwanted price risk to those willing to bear it. Exporters hedge currency risk with forwards, farmers hedge crop prices with futures, banks hedge interest rate risk with swaps, and investors insure portfolios with puts.
2. **Price discovery:** Futures and options prices reflect the market's collective expectations of future spot prices and volatility. Because trading is cheap and fast, new information is reflected first in derivative prices, which helps producers and consumers plan production and inventory.
3. **Operational advantages:** Derivatives have lower transaction costs than the spot market, higher liquidity, and make short selling easy (sell a futures or buy a put instead of borrowing shares). Large positions can be taken or reversed quickly.
4. **Market efficiency:** Arbitrage between spot and derivative markets (e.g., cash-and-carry, put–call parity) keeps prices of related assets in line, so mispricing is quickly eliminated and markets become more efficient.
5. **Speculation and leverage:** Speculators can take positions with a small initial outlay (margin or premium). They add liquidity and are the counterparties who absorb hedgers' risk.
6. **Completing the market:** Derivatives create payoff patterns (e.g., limited loss, insured portfolios, synthetic positions) that cannot be obtained easily with stocks and bonds alone, widening investors' choices.

Criticism/caution: Misuse (excessive speculation, poor understanding) can create large losses (e.g., Barings, 2008 crisis), so proper regulation and risk management are required. In Nepal, a formal derivative exchange is still developing, but forward contracts in foreign exchange and commodity markets are used.

Q12. An investor purchased a June put option with an exercise price of Rs 600 by paying a premium of Rs 50 and is holding it until June. Under what circumstances will the option be exercised? Under what circumstances will the holder of the option make a profit? With the help of a diagram, explain how the profit from a long position in the option depends on the stock price at the maturity of the option.

Solution:

Given: Long put; exercise price X = Rs 600; premium P = Rs 50; held till expiry (June).

Formula:

- Payoff of long put at expiry = max(0, X − S_T)
- Profit = max(0, X − S_T) − P
- Breakeven S_T = X − P

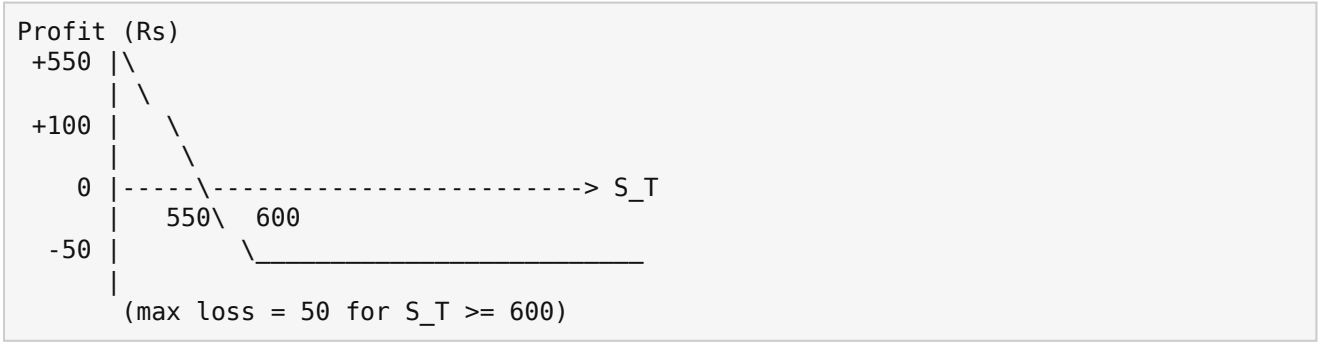
Step-by-step calculation:

1. **When exercised:** The put gives the right to sell at Rs 600. It is exercised only if S_T < 600 (in-the-money). If S_T ≥ 600 it expires worthless.
2. **When profitable:** Profit > 0 when X − S_T > P, i.e., 600 − S_T > 50 → **S_T < 550**. - Breakeven = 600 − 50 = **Rs 550**. - Between Rs 550 and Rs 600 the put is exercised (to recover part of the premium) but there is still a net loss.
3. **Maximum profit** = X − P = 600 − 50 = **Rs 550** (if S_T = 0).
4. **Maximum loss** = premium = **Rs 50** (if S_T ≥ 600).

Profit table:

S_T (Rs)	Exercise?	Payoff = max(0, 600 − S_T)	Premium	Profit (Rs)
450	Yes	150	50	+100
500	Yes	100	50	+50
550	Yes	50	50	0 (breakeven)
600	Indifferent	0	50	−50
650	No	0	50	−50
700	No	0	50	−50

Diagram (profit of long put vs stock price at maturity): The profit line starts at +550 on the vertical axis (S_T = 0), slopes downward at 45° (each Rs 1 rise in S_T reduces profit by Rs 1), crosses the horizontal axis at the breakeven price Rs 550, reaches −50 at the exercise price Rs 600 and then stays flat at −50 for all higher prices.



Answer / Interpretation: The put is exercised when S_T < Rs 600 and gives a profit when S_T < Rs 550. Profit rises by Rs 1 for each Rs 1 fall in the stock price below 600, with maximum profit Rs 550 and maximum loss Rs 50. A long put is a bearish strategy with limited loss and large (but limited) gain.

> 💡 **Exam Hint (remember):** Long put profit = max(0, X − S_T) − P; Breakeven = X − P; Max profit = X − P; Max loss = P. **Trap:** "exercised" (S_T < X) and "profit" (S_T < X − P) are different conditions — between 550 and 600 you exercise but still lose. **Shortcut:** for a put, everything is "X minus": BE = X − P, max gain = X − P.

Q13. An equity swap between the Employee Provident Fund (EPF) and Nabil Dealer is entered into. EPF pays floating interest payments based on SOFR, and Nabil pays a return on the NEPSE index. The payments are made quarterly, assuming a 30-day month and a 360-day year. The NEPSE index at the time of swap initiation was 2745.63, and the 90-day SOFR was 8 percent. Now, assume it is 50 days into the life of the swap. The NEPSE is now 2825.02, and the 40-day SOFR rate is 7.8 percent. Find the value of the swap today. Assume a notional principal of Rs 50,000,000.

Solution:

Given: NP = Rs 50,000,000; S_0 = 2745.63; S_t = 2825.02; initial 90-day SOFR L_0 = 8%; now day 50, so next payment in 40 days; 40-day SOFR = 7.8%. EPF pays floating (SOFR), receives NEPSE return. Nabil pays NEPSE return, receives SOFR.

Formula:

- Value of equity leg (per Re 1 notional) = S_t / S_0
- Value of floating leg (per Re 1) = (1 + L_0 × 90/360) × B_t(next payment)
- Discount factor B(40) = 1 / (1 + 0.078 × 40/360)
- Value to equity receiver (EPF) = NP × [S_t/S_0 − (1 + L_0 × 90/360) × B(40)]

Logic: Adding a notional to both legs does not change the value. The equity leg with notional is like holding NP rupees of the index, now worth NP × S_t/S_0. The floating leg resets to par (Re 1) right after the next payment, so today it is worth the PV of the already-known next payment plus notional.

Step-by-step calculation:

1. Index ratio: S_t / S_0 = 2825.02 / 2745.63 = **1.028915**
2. Discount factor: B(40) = 1 / (1 + 0.078 × 40/360) = 1 / (1 + 0.008667) = 1 / 1.008667 = **0.991408**
3. Next floating payment + notional (per Re 1) = 1 + 0.08 × 90/360 = 1 + 0.02 = 1.02
4. Floating leg value (per Re 1) = 1.02 × 0.991408 = **1.011236**
5. Equity leg value = 50,000,000 × 1.028915 = **Rs 51,445,752**
6. Floating leg value = 50,000,000 × 1.011236 = **Rs 50,561,798**
7. Value to EPF (receive equity, pay floating) = 51,445,752 − 50,561,798 = **+Rs 883,954**
8. Value to Nabil (receive floating, pay equity) = **−Rs 883,954**

Leg	Per Re 1	Value (Rs)
NEPSE (equity) leg	1.028915	51,445,752

Leg	Per Re 1	Value (Rs)
SOFR (floating) leg	1.011236	50,561,798
Net to EPF	0.017679	+883,954

Answer / Interpretation: The swap is worth **Rs 883,954 to EPF** (an asset) and **−Rs 883,954 to Nabil Dealer** (a liability). The NEPSE has risen 2.89% in 50 days, more than the PV of the floating payment, so the equity receiver gains.

> 💡 **Exam Hint (remember):** V(equity receiver) = NP × [S_t/S_0 − (1 + L_0 × m/360) × B(days to next payment)]. **Trap:** use the OLD rate L_0 (8%, already fixed) for the next payment and the NEW rate (7.8%, 40 days) only for discounting; don't use 90 days for discounting. **Shortcut:** floating leg = "next payment + 1, discounted once" because it resets to par.

Q14. People's Power Company (PCC), a Nepalese hydro-power company, purchased equipment from Baltimore Heavy Industry, a U. S. firm, for $5 million payable in one year. The current spot rate is Rs 140 per dollar, and the one-year forward rate is 130 yen per dollar. The annual interest rate is 5% in Nepal, and 8% in the U.S. PCC can also buy a one-year call option on the dollar at the strike price of Rs 125 per dollar for a premium of Rs 14 per dollar.

a. Assuming that the forward exchange rate is the best predictor of the future spot rate, compute the expected future rupee cost of meeting this obligation when the option hedge is used. b. At what future spot rate do you think PCC may be indifferent between the option and forward hedge?

Solution:

(The "130 yen per dollar" in the question is a typo for **Rs 130 per dollar**.)

Given: Payable = $5,000,000 in 1 year; spot S_0 = Rs 140/$; forward F = Rs 130/$; i(Nepal) = 5%; i(US) = 8%; call on $: X = Rs 125/$, premium c = Rs 14/$ (paid today).

Formula:

- Expected future spot E(S_T) = F (forward is unbiased predictor)
- Option hedge cost per $ at T = min(S_T, X) + c × (1 + i_Nepal)
- Forward hedge cost per $ = F
- Indifference: S_T + c(1 + i_Nepal) = F (for S_T ≤ X)

Step-by-step calculation:

(a) Expected cost with option hedge

1. E(S_T) = Rs 130/$ > X = Rs 125 → the call will be exercised; PCC buys dollars at Rs 125.
2. Premium paid today, carried to year-end at the Nepalese rate: 14 × 1.05 = **Rs 14.70/$**
3. Expected cost per $ = 125 + 14.70 = **Rs 139.70/$**
4. Total expected rupee cost = 5,000,000 × 139.70 = **Rs 698,500,000**

Comparison:

Strategy	Cost per $ (Rs)	Total cost (Rs)
Option hedge (expected)	139.70	698,500,000
Forward hedge (certain)	130.00	650,000,000
Unhedged (expected, S_T = 130)	130.00	650,000,000

(b) Indifference rate

- If S_T ≥ 125: option is exercised, cost = 125 + 14.70 = 139.70 > 130 → forward is always cheaper.
- If S_T < 125: option lapses, PCC buys at spot; cost = S_T + 14.70.
- Set equal to forward: S_T + 14.70 = 130 → **S_T = Rs 115.30/$**

Future spot S_T	Option hedge cost/$	Forward cost/$	Better
110	110 + 14.70 = 124.70	130	Option
115.30	130.00	130	Indifferent
120	134.70	130	Forward
125 and above	139.70 (capped)	130	Forward

Answer / Interpretation: (a) Expected rupee cost with the option hedge = **Rs 139.70/$ = Rs 698.5 million**. (b) PCC is indifferent at a future spot rate of **Rs 115.30/$**; if the dollar falls below Rs 115.30, the option hedge is cheaper; above it, the forward hedge is cheaper. If the forward is an unbiased predictor, the forward hedge (Rs 650 million, with no uncertainty) is the best choice.

💡 **Exam Hint (remember):** Option hedge cost (payable) = min(S_T, X) + premium × (1 + home rate). Indifference: S_T* = F − premium × (1 + r_home). **Trap:** don't forget to compound the premium (it is paid today, the obligation is paid in a year), and use the HOME (Nepal) rate because the premium is in rupees. **Shortcut:** Indifference = 130 − 14.70 = 115.30.

Q15. HDL stock is currently selling at Rs 900. A put option on HDL has an exercise price of Rs 1000 and matures in one year. The annually compounded interest rate is 8 percent. The quoted price of the put option is Rs 60. Identify the appropriate arbitrage opportunity and show the arbitrage profit for the stock price of Rs 800 and Rs 1200 after one year.

Solution:

Given: S_0 = Rs 900; X = Rs 1,000; T = 1 year; r = 8% (annual compounding); quoted put price P = Rs 60.

Formula:

- Lower bound of an American put: P ≥ max(0, X − S_0)
- Lower bound of a European put: P ≥ max(0, X/(1 + r)^T − S_0)

Step-by-step calculation:

1. Intrinsic value = X − S_0 = 1,000 − 900 = **Rs 100**
2. Lower bound (American put) = max(0, 100) = Rs 100

3. Quoted price Rs 60 < Rs 100 → **the put is underpriced** → arbitrage: **buy the put and buy the stock**.
4. Outlay today = 60 + 900 = **Rs 960**
5. Immediate exercise: deliver the stock at X = Rs 1,000 → riskless profit today = 1,000 − 960 = **Rs 40**.

If the position is instead held to expiry (protective put):

Action today	Cash flow today	S_T = 800	S_T = 1,200
Buy put	−60	Exercise: +(1,000 − 800) = +200	Expires: 0
Buy stock	−900	Stock worth +800	Sell stock +1,200
Total	**−960**	**+1,000**	**+1,200**
Arbitrage profit (vs outlay 960)		**1,000 − 960 = Rs 40**	**1,200 − 960 = Rs 240**

At S_T = 800 the investor exercises the put and receives Rs 1,000 → profit Rs 40. At S_T = 1,200 the put lapses and the stock is sold for Rs 1,200 → profit Rs 240. A minimum of Rs 40 is guaranteed in all cases, with extra upside.

Note (European bound and financing): European lower bound = 1,000/1.08 − 900 = 925.93 − 900 = **Rs 25.93**; the price 60 does not violate it. If the Rs 960 were borrowed at 8% (repay 960 × 1.08 = 1,036.80), the hold-to-maturity strategy would lose 1,000 − 1,036.80 = −36.80 in the Rs 800 case. Hence the pure riskless arbitrage relies on **immediate exercise of the American put** (profit Rs 40 now); without borrowing, the Rs 40 minimum profit is locked in either way.

Answer / Interpretation: The put is underpriced (60 < intrinsic value 100). Buy the put and the stock for Rs 960. Arbitrage profit = **Rs 40 at S_T = 800** and **Rs 240 at S_T = 1,200** (or Rs 40 immediately by early exercise).

> 💡 **Exam Hint (remember):** American put ≥ X − S; European put ≥ X/(1 + r)^T − S. If P < bound → BUY put + BUY stock (cheap insurance). **Trap:** check WHICH bound applies — the American bound (100) is violated, the European bound (25.93) is not. **Shortcut:** riskless profit = bound − price = 100 − 60 = 40.

Q16. The stock of Birat Company is currently selling for Rs 650. A 6-month call option on the stock has an exercise price of Rs 660. The standard deviation of returns on Birat stock has been estimated to be 50 percent. Currently, the 6-month T-bill rate is 10 percent. Using the BSM model, calculate the price of the call option on Birat.

Solution:

Given: S = Rs 650; X = Rs 660; σ = 0.50; r = 10% (continuously compounded); T = 6/12 = 0.5 year.

Formula:

- C = S × N(d1) − X × e^(−rT) × N(d2)
- d1 = [ln(S/X) + (r + σ²/2)T] / (σ√T)

- d2 = d1 − σ√T

Step-by-step calculation:

1. ln(S/X) = ln(650/660) = ln(0.984848) = **−0.015267**
2. σ²/2 = 0.25/2 = 0.125; (r + σ²/2)T = (0.10 + 0.125) × 0.5 = **0.1125**
3. σ√T = 0.5 × √0.5 = 0.5 × 0.707107 = **0.353553**
4. d1 = (−0.015267 + 0.1125) / 0.353553 = 0.097233 / 0.353553 = **0.2750**
5. d2 = 0.2750 − 0.3536 = **−0.0785**
6. N(d1) = N(0.2750) = **0.6083** - From the normal table: N(0.27) = 0.6064, N(0.28) = 0.6103 → N(0.275) = 0.6064 + 0.5 × (0.6103 − 0.6064) = 0.6083
7. N(d2) = N(−0.0785) = 1 − N(0.0785) = 1 − 0.5313 = **0.4687** - N(0.07) = 0.5279, N(0.08) = 0.5319 → N(0.0785) = 0.5279 + 0.85 × 0.0040 = 0.5313
8. e^(−rT) = e^(−0.10 × 0.5) = e^(−0.05) = **0.951229**
9. S × N(d1) = 650 × 0.6083 = **395.43**
10. X × e^(−rT) × N(d2) = 660 × 0.951229 × 0.4687 = 627.81 × 0.4687 = **294.26**
11. C = 395.43 − 294.26 = **Rs 101.17**

Answer / Interpretation: The BSM value of the Birat call is **≈ Rs 101.17 (≈ Rs 101)**. The option is slightly out-of-the-money (650 < 660), so its entire value is time value, which is high because volatility (50%) is high.

> 💡 **Exam Hint (remember):** C = S·N(d1) − X·e^(−rT)·N(d2); d1 = [ln(S/X) + (r + σ²/2)T]/(σ√T); d2 = d1 − σ√T. **Trap:** T in years (6 months = 0.5), σ² not σ in d1, and N(−d) = 1 − N(d). **Shortcut:** compute σ√T once (0.3536) and reuse it for both d1 and d2.

Q17. An investor enters into two long futures contracts on orange. Each contract has a contract size of 10,000 pounds. The current futures price is Rs 100 per pound. The initial margin is Rs 3,00,000 per contract, and the maintenance margin is Rs 1,50,000.

a. What price change would lead to a margin call? b. What happens if the margin call money is not deposited? c. Under what circumstances could Rs 1,00,000 be withdrawn from the margin account?

Solution:

Given: Long 2 contracts × 10,000 lb = 20,000 lb; F_0 = Rs 100/lb; initial margin = 3,00,000 × 2 = Rs 6,00,000; maintenance margin = 1,50,000 × 2 = Rs 3,00,000.

Formula:

- Gain/loss on margin account = (F_t − F_0) × total quantity
- Margin call when balance < maintenance margin → loss > Initial − Maintenance
- Withdrawable amount = Balance − Initial margin

Step-by-step calculation:

(a) Margin call

1. Loss that triggers a call = 6,00,000 − 3,00,000 = **Rs 3,00,000**
2. Price fall = 3,00,000 / 20,000 = **Rs 15 per lb**

3. Futures price at which call occurs = 100 − 15 = **Rs 85 per lb** — a margin call is received if the price falls by more than Rs 15 (below Rs 85). The investor must then deposit variation margin to bring the balance back to the initial margin (Rs 6,00,000).

Futures price (Rs/lb)	Gain/Loss (Rs)	Balance (Rs)	Status
100	0	6,00,000	Initial
90	−2,00,000	4,00,000	No call
85	−3,00,000	3,00,000	At maintenance level (call if below)
80	−4,00,000	2,00,000	Margin call: deposit 4,00,000

(b) If the margin call is not met: the broker/clearinghouse **closes out (liquidates) the position** by taking an offsetting short position in the futures. The loss is realised and charged to the margin account; any remaining balance is returned to the investor (any deficit must be paid by the investor).

(c) Withdrawal of Rs 1,00,000

1. Any balance above the initial margin can be withdrawn. Need balance = 6,00,000 + 1,00,000 = 7,00,000 → gain = Rs 1,00,000
2. Price rise = 1,00,000 / 20,000 = **Rs 5 per lb**
3. Futures price = 100 + 5 = **Rs 105 per lb** — Rs 1,00,000 can be withdrawn when the futures price rises to Rs 105 or more.

Answer / Interpretation: (a) A fall of Rs 15/lb (price below Rs 85) triggers a margin call. (b) The position is closed out. (c) Rs 1,00,000 can be withdrawn if the futures price rises to Rs 105 (gain of Rs 5/lb).

> 💡 **Exam Hint (remember):** Price change for margin call = (Initial − Maintenance) ÷ total units; withdrawal price change = Amount ÷ total units. **Trap:** multiply margins and units by the NUMBER of contracts (2), and after a call you top up to the INITIAL margin, not the maintenance margin. **Shortcut:** per contract: 1,50,000/10,000 = Rs 15 — the number of contracts cancels out.

Group C — Long-answer Questions (Any three, 3 × 10 = 30)

Q18. NRIC stock is currently trading at Rs 1200 per share. At-the-money options on MRIC with 180 days to expiration are actively traded in the option exchange. You want to use the binomial model to value this option and expect that either the stock price increases by 25 percent or it decreases by 20 percent at the end of six months. You can borrow money at 8 percent per annum. Assume a 360-day year.

a. Calculate the fair price of the call option today. b. Calculate the arbitrage return in percentage if the actual price of this call option in the market is Rs 140.

Solution:

Given: S = Rs 1,200; X = Rs 1,200 (at-the-money); u = 1.25; d = 0.80; r = 8% p.a. → per period (180 days) r = 0.08 × 180/360 = 4% → (1 + r) = 1.04; market call price = Rs 140.

Formula:

- Su = S × u; Sd = S × d; Cu = max(0, Su − X); Cd = max(0, Sd − X)
- Risk-neutral probability p = [(1 + r) − d] / (u − d)
- C = [p × Cu + (1 − p) × Cd] / (1 + r)
- Hedge ratio h = (Cu − Cd) / (Su − Sd)

Step-by-step calculation:

(a) Fair price

1. Su = 1,200 × 1.25 = **Rs 1,500**; Sd = 1,200 × 0.80 = **Rs 960**
2. Cu = max(0, 1,500 − 1,200) = **Rs 300**; Cd = max(0, 960 − 1,200) = **0**
3. p = (1.04 − 0.80) / (1.25 − 0.80) = 0.24 / 0.45 = **0.5333**; 1 − p = 0.4667
4. C = (0.5333 × 300 + 0.4667 × 0) / 1.04 = 160 / 1.04 = **Rs 153.85**
5. Hedge ratio h = (300 − 0) / (1,500 − 960) = 300 / 540 = **0.5556 shares per call**

Check (replication): Hedge portfolio = buy 0.5556 shares, write 1 call. Value at T: up = 0.5556 × 1,500 − 300 = 533.33; down = 0.5556 × 960 − 0 = 533.33 (riskless). PV = 533.33/1.04 = 512.82. So 0.5556 × 1,200 − C = 512.82 → C = 666.67 − 512.82 = 153.85 ✓

(b) Arbitrage with market price Rs 140

Market price 140 < fair value 153.85 → **call is underpriced → buy the call and short h = 0.5556 shares**, lend the proceeds at 4%.

Per call:

Action	Today	Up (S_T = 1,500)	Down (S_T = 960)
Short 0.5556 shares	+666.67	−833.33	−533.33
Buy 1 call	−140.00	+300.00	0
Lend net proceeds at 4%	−526.67	+547.73	+547.73
Net	**0**	**+14.40**	**+14.40**

1. Net inflow today = 666.67 − 140 = **526.67**
2. Invested at 4%: 526.67 × 1.04 = **547.73**
3. Net outflow at expiry (either state) = **533.33** (up: 833.33 − 300; down: 533.33 − 0)
4. Riskless profit = 547.73 − 533.33 = **Rs 14.40 per call**
5. Arbitrage return = 14.40 / 526.67 = **2.73% per six months ≈ 5.47% per annum** (over and above the risk-free return).

Alternative view: shorting the hedge portfolio means effectively borrowing 526.67 today and repaying 533.33 later, i.e., borrowing at 533.33/526.67 − 1 = **1.27% per half year (2.53% p.a.)** versus the market rate of 4% (8% p.a.) — a saving of 2.73% per period.

Scaled example (1,000 calls, 556 shares):

Action	Today (Rs)	Up (Rs)	Down (Rs)
Short 556 shares	+667,200	−834,000	−533,760
Buy 1,000 calls	−140,000	+300,000	0
Lend 527,200 at 4%	−527,200	+548,288	+548,288
Net	**0**	**+14,288**	**+14,528**

Answer / Interpretation: (a) Fair call price = **Rs 153.85** (hedge ratio 0.5556). (b) At Rs 140 the call is underpriced; buying calls and shorting 0.5556 shares per call earns a

riskless **Rs 14.40 per call = 2.73% per six months (≈ 5.47% p.a.) excess return**. Arbitrage buying will push the call price up toward Rs 153.85.

> 💡 **Exam Hint (remember):** p = (1 + r − d)/(u − d); C = [pCu + (1 − p)Cd]/(1 + r); h = (Cu − Cd)/(Su − Sd). **Trap:** r must be for the period (8% × 180/360 = 4%), NOT 8%; and if C(market) < C(fair) you BUY the call and SHORT h shares (reverse if overpriced). **Shortcut:** arbitrage profit per call at T = (C_fair − C_market) × (1 + r) = 13.85 × 1.04 = 14.40.

Q19. A one-year swap with quarterly payment and notional principal of Rs 100 million is entered between Citizen Investment Trust (CIT) and Global Bank. Payments will be based on a 30-day month and a 360-day year. The CIT agreed to make interest payments based on the 90-day T-bill rate, and Global Bank agreed to make interest payments based on the 90-day SOFR minus a spread.

a. Find the spread that should be deducted from the SOFR rate on each payment date. The term structures of interest rates are given below.

Term	SOFR rate	T-bill rate
90 days	7.50%	5.50%
180 days	7.95%	5.87%
270 days	8.52%	6.30%
360 days	9.00%	6.90%

b. Assume it is 60 days into the life of the swap. The new term structures of interest rates are as follows. Find the value of the swap for both counterparties.

Term	SOFR rate	T-bill rate
30 days	8.00%	6.00%
120 days	8.45%	6.47%
210 days	9.00%	6.95%
300 days	9.55%	7.40%

Solution:

Given: NP = Rs 100 million; quarterly payments (90 days); 30/360. CIT pays T-bill, receives SOFR − spread; Global Bank pays SOFR − spread, receives T-bill.

Formula:

- Discount factor B(t) = 1 / (1 + r × t/360)
- Fixed rate of a plain-vanilla swap (per quarter) = [1 − B(360)] / ΣB; annual = × 360/90
- Basis swap spread = Fixed rate (SOFR swap) − Fixed rate (T-bill swap)
- Value of floating leg (with notional) = (1 + L_0 × 90/360) × B(days to next payment)
- Value of spread leg = NP × spread × 90/360 × ΣB (remaining)

Logic: A basis swap = (pay T-bill, receive fixed) + (pay fixed, receive SOFR). For zero initial value, the spread must equal the difference between the fixed rates of the two plain-vanilla swaps.

Step-by-step calculation:

(a) Spread

SOFR discount factors:

Days	Rate	Calculation	B
90	7.50%	1/(1 + 0.075 × 90/360) = 1/1.01875	0.98160
180	7.95%	1/(1 + 0.0795 × 180/360) = 1/1.03975	0.96177
270	8.52%	1/(1 + 0.0852 × 270/360) = 1/1.06390	0.93994
360	9.00%	1/(1 + 0.09 × 360/360) = 1/1.09000	0.91743
		Σ	**3.80073**

Fixed rate (SOFR swap) = (1 − 0.91743)/3.80073 = 0.021724 per quarter → × 4 = **8.69% p.a.**

T-bill discount factors:

Days	Rate	Calculation	B
90	5.50%	1/(1 + 0.055 × 90/360) = 1/1.01375	0.98644
180	5.87%	1/(1 + 0.0587 × 180/360) = 1/1.02935	0.97149
270	6.30%	1/(1 + 0.063 × 270/360) = 1/1.04725	0.95488
360	6.90%	1/(1 + 0.069 × 360/360) = 1/1.06900	0.93545
		Σ	**3.84826**

Fixed rate (T-bill swap) = (1 − 0.93545)/3.84826 = 0.016773 per quarter → × 4 = **6.71% p.a.**

Spread = 8.69% − 6.71% = **1.98% p.a. (≈ 198 basis points)**, deducted from SOFR on each payment date. So Global Bank pays (SOFR − 1.98%) and CIT pays T-bill.

(b) Value 60 days into the swap (next payment in 30 days, then 120, 210, 300 days)

New discount factors:

Days	SOFR B	T-bill B
30	1/(1 + 0.08 × 30/360) = 0.99338	1/(1 + 0.06 × 30/360) = 0.99502
120	1/(1 + 0.0845 × 120/360) = 0.97260	1/(1 + 0.0647 × 120/360) = 0.97889
210	1/(1 + 0.09 × 210/360) = 0.95012	1/(1 + 0.0695 × 210/360) = 0.96104
300	1/(1 + 0.0955 × 300/360) = 0.92628	1/(1 + 0.074 × 300/360) = 0.94192
Σ	3.84238	—

1. **SOFR floating leg** (next payment fixed at initial 90-day SOFR 7.5%): 100,000,000 × (1 + 0.075 × 90/360) × 0.99338 = 100,000,000 × 1.01875 × 0.99338 = **Rs 101,200,331**
2. **Spread leg** (fixed payment each quarter): 100,000,000 × 0.0198063 × 90/360 = **Rs 495,158** per quarter; PV = 495,158 × 3.84238 = **Rs 1,902,587**
3. Net value of SOFR − spread leg = 101,200,331 − 1,902,587 = **Rs 99,297,744**
4. **T-bill leg** (next payment fixed at initial 90-day T-bill 5.5%): 100,000,000 × (1 + 0.055 × 90/360) × 0.99502 = 100,000,000 × 1.01375 × 0.99502 = **Rs 100,870,647**
5. Value to CIT (receives SOFR − spread, pays T-bill) = 99,297,744 − 100,870,647 = **−Rs 1,572,903**

6. Value to Global Bank (receives T-bill, pays SOFR − spread) = **+Rs 1,572,903**

Answer / Interpretation: (a) Spread = **1.98% (≈ 198 bp)** deducted from SOFR. (b) After 60 days, the swap is worth **−Rs 1,572,903 to CIT** and **+Rs 1,572,903 to Global Bank**. Both floating legs reset to par at each payment date, so each is valued as the PV of the next known payment plus notional; the spread payments are a fixed annuity discounted at the SOFR factors.

> 💡 **Exam Hint (remember):** Swap fixed rate = (1 − B_last)/ΣB × (360/90); Basis spread = Fixed(SOFR) − Fixed(T-bill); Floating leg value = (1 + L_0 × 90/360) × B(next). **Trap:** the next payment uses the OLD (initial) 90-day rates (7.5% and 5.5%), not the new 30-day rates; days to discount are 30, 120, 210, 300 (not 90, 180…). **Shortcut:** compute both columns of discount factors in one table and reuse ΣB.

Q20. A stock is expected to pay a dividend of Rs 10 per share in two months and five months from today. The stock price is Rs 500, and the risk-free interest rate is 8% p.a. with continuous compounding for all maturities. An investor has just taken a short position in a six-month forward contract on the stock.

a. What are the forward price and the initial value of the forward contract? b. Show the arbitrage process if the market price of the forward is Rs 515. b. Three months later, the stock price is Rs 480, and the risk-free rate of interest is still 8% p.a. What are the forward price and the value of the short position in the forward contract?

Solution:

Given: S_0 = Rs 500; D = Rs 10 at t = 2/12 and 5/12; r = 8% (continuous); T = 6/12 = 0.5; market forward = Rs 515; after 3 months S = Rs 480.

Formula:

- PV of dividends I = Σ D × e^(−r t)
- F_0 = (S_0 − I) × e^(rT)
- Value of short forward at time t = (K − F_t) × e^(−r(T − t)) = K e^(−r(T−t)) − (S_t − I_t)

Step-by-step calculation:

(a) Forward price and initial value

1. I = 10e^(−0.08 × 2/12) + 10e^(−0.08 × 5/12) = 10e^(−0.013333) + 10e^(−0.033333) = 9.8676 + 9.6722 = **19.5397**
2. S_0 − I = 500 − 19.5397 = 480.4603
3. e^(0.08 × 0.5) = e^(0.04) = 1.040811
4. F_0 = 480.4603 × 1.040811 = **Rs 500.07**
5. Initial value of the forward = **0** (delivery price is set equal to the forward price, so no money changes hands).

(b) Arbitrage at market forward Rs 515

Market forward 515 > fair 500.07 → forward is overpriced → **cash-and-carry arbitrage: sell the forward, borrow, buy the stock**.

Time	Action	Cash flow (Rs)
Today	Short forward at 515	0
Today	Borrow 500 for 6 months at 8% cc	+500.00
Today	Buy 1 share	−500.00
2 months	Receive dividend 10, invest for 4 months: 10e^(0.08 × 4/12)	→ 10.2702 at T
5 months	Receive dividend 10, invest for 1 month: 10e^(0.08 × 1/12)	→ 10.0669 at T
6 months	Deliver share under forward	+515.00
6 months	Receive invested dividends	+20.34
6 months	Repay loan 500e^(0.04)	−520.41
	Arbitrage profit at T	**+14.93**

Profit = 515 + 20.34 − 520.41 = **Rs 14.93** (= 515 − 500.07). Selling forwards and buying stock will push the forward price down toward Rs 500.07.

(c) After 3 months (T − t = 3 months = 0.25 year)

1. Only one dividend remains (at month 5, i.e., 2 months from now): I = 10e^(−0.08 × 2/12) = **9.8676**
2. S − I = 480 − 9.8676 = 470.1324
3. e^(0.08 × 0.25) = e^(0.02) = 1.020201
4. New forward price F_t = 470.1324 × 1.020201 = **Rs 479.63**
5. Value of short forward = (K − F_t) × e^(−0.08 × 0.25) = (500.07 − 479.63) × 0.980199 = 20.44 × 0.980199 = **Rs 20.03** - Check: K e^(−r(T−t)) − (S − I) = 500.07 × 0.980199 − 470.13 = 490.17 − 470.13 = 20.03 ✓

Answer / Interpretation: (a) F_0 = **Rs 500.07**; initial value = 0. (b) Forward at 515 is overpriced; cash-and-carry gives a riskless profit of **Rs 14.93** per share. (c) After 3 months, forward price = **Rs 479.63**; value of the short position = **+Rs 20.03** (the short gains because the stock price fell).

> 💡 **Exam Hint (remember):** F_0 = (S_0 − PV of dividends) × e^(rT); value of short = (K − F_t)e^(−r(T−t)); value of long = (F_t − K)e^(−r(T−t)). **Trap:** after 3 months, the 2-month dividend has ALREADY been paid — include only the remaining one, discounted for 2 months (not 5). **Shortcut:** arbitrage profit = market forward − fair forward (515 − 500.07 = 14.93).

Q21. Explain the margin rule and the margin operation process by a clearinghouse in option trading.

Solution:

Meaning of margin in options

Margin is a performance bond (good-faith deposit) that a trader must keep with the broker/clearinghouse to ensure that contractual obligations will be met. In options, the right belongs to the buyer and the obligation to the writer, so margin rules differ for buyers and writers.

Margin rules

1. **Option buyers:** The buyer has a right but no obligation. Therefore, the buyer pays the **full premium** upfront (no buying on margin for short-term options; long-term options may be partly margined). The maximum loss is the premium, so no further margin is needed.
2. **Covered call writers:** A writer who owns the underlying stock (covered call) needs **no additional margin**; the shares are deposited/held with the broker and can be delivered if exercised.
3. **Uncovered (naked) call writers:** Because the potential loss is unlimited, the writer must deposit: - Margin = 100% of the premium received + 20% of the underlying stock value − amount by which the option is out-of-the-money, - subject to a minimum of premium + 10% of the stock value. - Example: S = 100, X = 105, premium = 4 for 100 shares: 400 + 0.20 × 10,000 − 500 = 1,900; minimum = 400 + 1,000 = 1,400 → margin = Rs 1,900.
4. **Uncovered put writers:** Margin = premium + 20% of stock value − out-of-the-money amount, minimum premium + 10% of the exercise price. A cash-secured put (cash equal to the strike deposited) needs no additional margin.
5. **Spreads and combinations:** For spreads, the margin is based on the maximum possible loss (difference between strikes minus net premium), since the long leg protects the short leg.
6. Margins are **recomputed daily** as prices change; if the margin is insufficient, a margin call is issued.

Margin operation process by the clearinghouse

1. **Novation / guarantee:** The clearinghouse (e.g., OCC in the USA) becomes the buyer to every seller and the seller to every buyer. It guarantees the performance of every contract, eliminating counterparty default risk.
2. **Clearing members:** Only clearing members deal directly with the clearinghouse. Brokers who are not members clear through members. Customers post margin with brokers, brokers with clearing members, and clearing members with the clearinghouse.
3. **Daily collection and marking-to-market:** The clearinghouse values all short positions daily and collects additional margin from clearing members whose positions have lost value; members in turn collect from customers.
4. **Margin calls and liquidation:** If a member or customer fails to meet a call, the position is closed out and the margin used to cover losses.
5. **Exercise and assignment:** When a buyer exercises, the clearinghouse assigns the exercise **randomly** to a clearing member with a short position, who then assigns it to one of its customer writers (random or FIFO).
6. **Clearing/guarantee fund and default waterfall:** If a member defaults, losses are met first from the defaulter's margin, then its clearing fund contribution, then the clearinghouse's own capital and the contributions of other members.

Nepalese context: NEPSE does not yet have an organised options exchange; SEBON has issued commodity/derivative regulations and the market is at an early stage, so these margin practices are drawn from international exchanges.

Conclusion: Margin rules protect the system from writers' default, while the clearinghouse's daily margining, guarantee and assignment process makes option markets safe and liquid.

Group D — Comprehensive / Case Analysis (20)

Q22. Krishna Pauroti Company is an exchange-listed company that produces and sells different brands of bread throughout Nepal. The company's products gained popularity in a short period. The company's earnings and stock price are growing rapidly. Since its stock is actively traded on the stock exchange, it is also listed on the options exchange. Call and put options on Krishna stock, which is also actively trading. The company has announced that it is planning to expand its capacity by 100 percent within six months to meet the increased demand. Yesterday, a case against the Company was filed in court, stating that a popular brand of the Company's bread is unhealthy and the company is knowingly ignoring public health. The court had issued a stay order to the company to stop the production and sale of that particular brand of bread until the court gave a final verdict. Today's trade volume of Krishna stock has become very low. The stock closed at Rs 240 per share today. Call and put options on Krishna with an exercise price of Rs 240 and three months to expiration closed at Rs 40 and Rs 20, respectively. Stock traders in the market are waiting for the court verdict. Special courts handle such cases, and usually, the verdict comes within three months. The risk-free rate of interest is currently 6 percent per annum. You are considering investing Rs 240,000 in Krishna stock, an option on Krishna stock, or a combination of both.

a. If you believe the court verdict will favor the company and want to speculate on the option, which option do you buy? Why? How much will you earn/lose if the stock price after three months increases to Rs 350? What will be your annual rate of return? b. If you believe that the court verdict will come against the company and want to speculate on the option, which option do you buy? Why? How much will you earn/lose if the stock price after three months decreases to Rs 250? What will be your annual rate of return? Compare the risk and return of your investment with that of your stock investment. c. Assume you cannot guess whether the court verdict may come in favor of or against the company. Is a long straddle beneficial in such conditions? Why? Calculate gain/loss from long straddle for expiration stock prices of Rs 160, Rs 180, … Rs 320. After three months, calculate the maximum possible gain, maximum possible loss, and breakeven stock price(s) for your strategy. d. Considering the uncertainty of the verdict, you want to ensure your portfolio of stocks using put options. How many stocks and put options do you need to hold? What will be the ensured value of your portfolio? What will be the value of the ensured portfolio if the verdict came in favor of the company, and the stock price increased to Rs 350? Also, calculate the cost of insurance and the upside capture. [4+4+6+6]

Solution:

Given: S_0 = Rs 240; call (X = 240) C = Rs 40; put (X = 240) P = Rs 20; T = 3 months; r_f = 6% p.a.; investment = Rs 240,000.

Formula:

- Call profit = $\max(0, S_T - X) - C$; Put profit = $\max(0, X - S_T) - P$

- Straddle profit = |S_T − X| − (C + P); breakevens = X ± (C + P)
- Return = Profit / Investment; annual (simple) = 3-month return × 12/3
- Protective put: units = Investment / (S + P); floor = units × X; upside capture = insured value / uninsured value

Step-by-step calculation:

(a) Favourable verdict → buy CALL options

Why: a favourable verdict will push the price up. A call gives leveraged, unlimited upside with loss limited to the premium; a put would lose.

1. Number of calls = 240,000 / 40 = **6,000 calls**
2. Payoff at S_T = 350: (350 − 240) × 6,000 = 110 × 6,000 = **Rs 660,000**
3. Profit = 660,000 − 240,000 = **Rs 420,000**
4. 3-month return = 420,000 / 240,000 = **175%**
5. Annual rate of return = 175% × 12/3 = **700%** (simple); compounded = (1 + 1.75)^4 − 1 = (2.75)^4 − 1 ≈ 5,620%

Comparison: buying 1,000 shares (240,000/240) → profit = (350 − 240) × 1,000 = Rs 110,000 = 45.83% (183.3% annualised).

(b) Unfavourable verdict → buy PUT options

Why: an adverse verdict will push the price down. A put gains as the price falls, with loss limited to the premium (safer than short selling, which has unlimited loss and is not easy in Nepal).

1. Number of puts = 240,000 / 20 = **12,000 puts**

The question says the price "decreases to Rs 250", but Rs 250 is above Rs 240. Both readings are shown.

(i) As stated, S_T = Rs 250:

- Put payoff = max(0, 240 − 250) = 0 → puts expire worthless
- Loss = **−Rs 240,000** → return = −100% (−400% annualised, simple)
- Stock investment (1,000 shares): gain = (250 − 240) × 1,000 = +Rs 10,000 → +4.17% (16.67% annualised)

(ii) If the intended price is Rs 150:

- Put payoff = (240 − 150) × 12,000 = **Rs 1,080,000**
- Profit = 1,080,000 − 240,000 = **Rs 840,000** → return = 350% (**1,400% annualised**)
- Stock investment: loss = (150 − 240) × 1,000 = −Rs 90,000 → −37.5% (−150% annualised)

Case	Puts (12,000)	Return	Stock (1,000 shares)	Return
S_T = 250	−240,000	−100% (−400% p.a.)	+10,000	+4.17% (16.67% p.a.)
S_T = 150	+840,000	+350% (1,400% p.a.)	−90,000	−37.5% (−150% p.a.)

Risk–return comparison: options give much higher (leveraged) returns when the forecast is correct, but the whole investment (100%) can be lost if the price does not move as

expected. The stock investment gives smaller returns, but losses are only partial — lower risk, lower return.

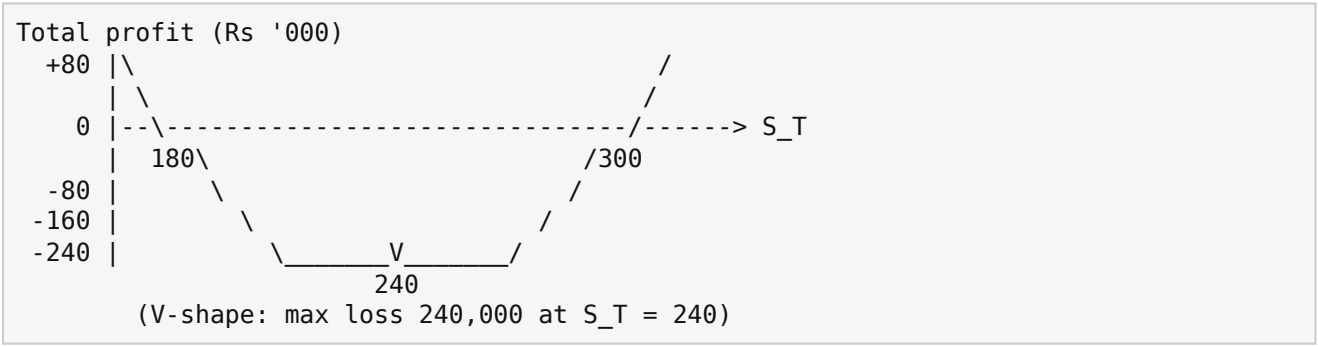
(c) Long straddle (buy 1 call + 1 put, X = 240)

Why: the verdict will cause a large price move but the direction is unknown. A long straddle profits from a big move in either direction; loss is limited to the total premium. So it is beneficial here.

1. Cost per straddle = 40 + 20 = Rs 60; number = 240,000 / 60 = **4,000 straddles**
2. Profit per straddle = |S_T − 240| − 60

S_T (Rs)	Call payoff	Put payoff	Total payoff	Profit per straddle	Total profit (× 4,000) (Rs)
160	0	80	80	20	80,000
180	0	60	60	0	0
200	0	40	40	−20	−80,000
220	0	20	20	−40	−160,000
240	0	0	0	−60	−240,000
260	20	0	20	−40	−160,000
280	40	0	40	−20	−80,000
300	60	0	60	0	0
320	80	0	80	20	80,000

3. Maximum loss = total premium = 60 × 4,000 = **Rs 240,000** (at S_T = 240)
4. Maximum gain = **unlimited** on the upside; on the downside maximum = (240 − 60) × 4,000 = **Rs 720,000** (if S_T = 0)
5. Breakeven prices = 240 − 60 = **Rs 180** and 240 + 60 = **Rs 300**



(d) Portfolio insurance with protective puts

1. One put per share; cost per unit = 240 + 20 = Rs 260
2. Units = 240,000 / 260 = 923.08 → **923 shares and 923 puts**; cost = 923 × 260 = Rs 239,980 (Rs 20 left as cash)
3. Ensured (floor) value = 923 × 240 = **Rs 221,520** (+ Rs 20 cash) — the minimum value whatever the verdict
4. If S_T = 350: puts expire worthless; value = 923 × 350 = **Rs 323,050**
5. Cost of insurance = 923 × 20 = **Rs 18,460** (= 20/240 = 8.33% of the stock value)
6. Uninsured portfolio (1,000 shares) at 350 = 1,000 × 350 = Rs 350,000
7. Upside capture = 323,050 / 350,000 = **92.3%**

Upside capture is the proportion of the uninsured portfolio's value that the insured portfolio retains in a rising market; it is below 100% because part of the money was spent on put premiums (fewer shares are held).

Answer / Interpretation:

- (a) Buy 6,000 calls; profit **Rs 420,000** (175% in 3 months; **700% p.a.**).
- (b) Buy 12,000 puts; at Rs 250 the puts expire worthless → loss **Rs 240,000** (−100%, −400% p.a.) vs stock gain Rs 10,000; if the price falls to Rs 150, profit Rs 840,000 (350%, 1,400% p.a.) vs stock loss Rs 90,000.
- (c) A long straddle suits the uncertain verdict: 4,000 straddles; max loss **Rs 240,000**, max gain unlimited (Rs 720,000 on the downside); breakevens **Rs 180 and Rs 300**.
- (d) Hold **923 shares + 923 puts**; floor **Rs 221,520**; value at 350 = **Rs 323,050**; insurance cost **Rs 18,460**; upside capture **92.3%**.

> 💡 **Exam Hint (remember):** Number of options = Investment ÷ premium; straddle BE = X ± (C + P); protective put units = Investment ÷ (S + P); floor = units × X; upside capture = insured ÷ uninsured value. **Trap:** annualise a 3-month return by × 4 (simple), and in (d) divide by (S + P), not by S. **Shortcut:** straddle profit changes Rs 20 × 4,000 = Rs 80,000 for every Rs 20 step in S_T — fill the table by adding/subtracting 80,000.